MAT1001

Calculus I

Homework Exercises

FALL 2026

565 assigned exercises · 48 sections

Selected from Thomas' Calculus in SI Units, 14th Edition.
Assignment scope follows MAT1001 Course Details 2026.

Shared directions precede the relevant selected exercises.
Original exercise numbers and necessary diagrams are retained.
Only assigned subparts are included where specified.

Sections follow the order in the course assignment list:
7.8 before 7.6; 7.4 after 16.1.

SF Pro Text & Display

Question collection • No solutions

README

INSTRUCTION

本材料分为两个文件，其一是题目集，收入了所有Syllabus上的题目（下文中称为Questions），其二是答案（下文中称为Answers），目前只包含5.1以后的章节。

1. Questions和Answers的目录均有超链接，点击可以直达对应章节。
2. Questions中，按照Syllabus把Thomas' Calculus in SI Units, 14th Edition中所有的题目按照Section提取出来了，包括必要的公式和配图，经核对，所有MAT1001的班级的题目都是一样的。排版上可能不算特别美观，但是对完成作业影响不大。蓝色标题是题目类型，共题头的题目会把总题目放在几道题的最前面。
3. 在Answers中，对于每一类题型都标注了课本上对应的例题，供大家类比学习。每道题的答案都包括了“中文思路讲解”和“完整英文答案”。

PREFACE

选修MAT1001的各位大家好啊！

开学后不久，本人在做MAT1001作业的时候，苦于要不断地翻看Syllabus，在教材上确定题号，然后再在goodnotes上作答。

于是萌生了“把所有题目提取成一份精华版的想法”，于是使用Chatgpt把所有题目提取成一份markdown文档，导入到Obsidian中。

当时效果还是不错的，而且这个举动也节约了我大量的时间，让我有更多的时间做自己想做的事情。

近期写完作业核对的时候，偶然发现了当时文档存在一些错误，同时也发现好多同学还是像我当时一样对着教材一题一题写作业。

于是本着开源的精神、以及希望能够让大家节省出更多时间做自己真的想做的事情，而不是把时间花在这样无意义的核对题目题号的举动上，我打算把这份“MAT1001 Homework Collection”开源给大家。

本人以及本人的舍友对这份文档进行了初步的核对，但是因为题量太大，所以难免还是会有出错的情况，希望大家见谅，也积极欢迎大家发现错误后向本人反馈，进行修改。

最后，祝愿大家都能把时间花在感兴趣的事情上！也祝愿大家MAT1001取得一个好的成绩！

By Howard in CHUKSZ

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2.1 Rates of Change and Tangent Lines to Curves

Assigned: 1, 7, 8, 19, 20

Thomas' Calculus, 14th SI edition • Section 2.1 • textbook pp. 61–63

Exercise 1: Average Rates of Change

Directions for Exercises 1–6.
In Exercises 1–6, find the average rate of change of the function over the given interval or intervals.

Exercise 1

$f(x) = x^3 + 1$

a. $[2, 3]$ **b.** $[-1, 1]$

Exercises 7, 8: Slope of a Curve at a Point

Directions for Exercises 7–18.
In Exercises 7–18, use the method in Example 3 to find **(a)** the slope of the curve at the given point P, and **(b)** an equation of the tangent line at P.

Exercise 7

$y = x^2 - 5, \quad P(2, -1)$

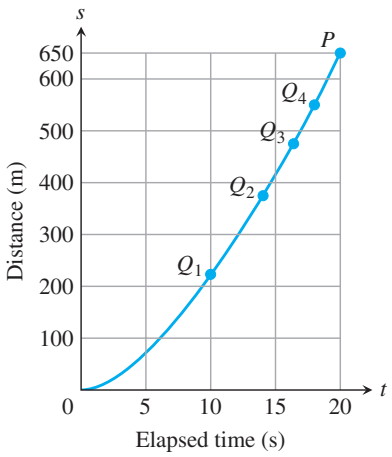
Exercise 8

$y = 7 - x^2, \quad P(2, 3)$

Exercises 19, 20: Instantaneous Rates of Change

Exercise 19

Speed of a car The accompanying figure shows the time-to-distance graph for a sports car accelerating from a standstill.



a. Estimate the slopes of secant lines PQ_1, PQ_2, PQ_3, and PQ_4, arranging them in order in a table like the one in Figure 2.6. What are the appropriate units for these slopes?

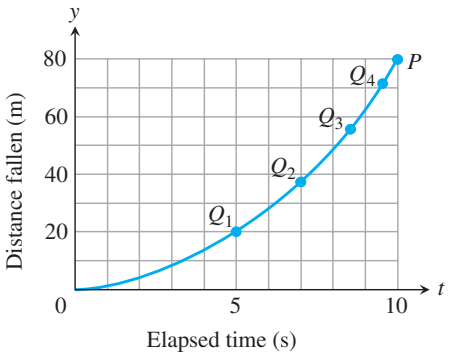
b. Then estimate the car's speed at time $t = 20$ s.

Exercise 20

The accompanying figure shows the plot of distance fallen versus time for an object that fell from the lunar landing module a distance 80 m to the surface of the moon.

a. Estimate the slopes of the secant lines PQ_1, PQ_2, PQ_3, and PQ_4, arranging them in a table like the one in Figure 2.6.

b. About how fast was the object going when it hit the surface?



2.2 Limit of a Function and Limit Laws

Assigned: 1, 4, 9, 10, 18, 19, 23, 28, 32, 40, 44, 45, 54, 57, 65, 75, 77, 80 • Specified subparts are marked below.

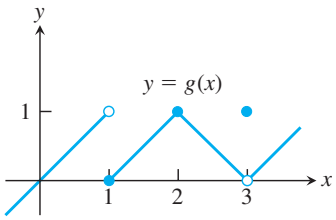
Thomas' Calculus, 14th SI edition • Section 2.2 • textbook pp. 71–74

Exercises 1, 4: Limits from Graphs

Exercise 1

For the function $g(x)$ graphed here, find the following limits or explain why they do not exist.

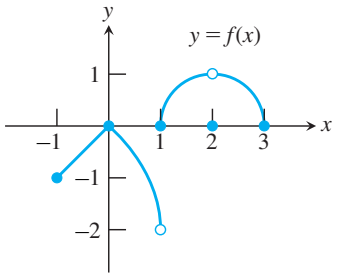
a. $\lim_{x\to 1} g(x)$ **b.** $\lim_{x\to 2} g(x)$ **c.** $\lim_{x\to 3} g(x)$ **d.** $\lim_{x\to 2.5} g(x)$



Exercise 4 • (a)-(e) only

Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

a. $\lim_{x\to 2} f(x)$ does not exist.
b. $\lim_{x\to 2} f(x) = 2$
c. $\lim_{x\to 1} f(x)$ does not exist.
d. $\lim_{x\to c} f(x)$ exists at every point c in $(-1, 1)$.
e. $\lim_{x\to c} f(x)$ exists at every point c in $(1, 3)$.



Exercises 9, 10: Existence of Limits

Exercise 9

If $\lim_{x\to 1} f(x) = 5$, must f be defined at $x = 1$? If it is, must $f(1) = 5$? Can we conclude *anything* about the values of f at $x = 1$? Explain.

Exercise 10

If $f(1) = 5$, must $\lim_{x\to 1} f(x)$ exist? If it does, then must $\lim_{x\to 1} f(x) = 5$? Can we conclude *anything* about $\lim_{x\to 1} f(x)$? Explain.

Exercises 18, 19, 23, 28, 32, 40, 44, 45: Calculating Limits

Directions for Exercises 11–22.
Find the limits in Exercises 11–22.

Exercise 18

$$\lim_{y\to 2} \frac{y + 2}{y^2 + 5y + 6}$$

Exercise 19

$$\lim_{y\to -3} (5 - y)^{4/3}$$

Directions for Exercises 23–42.
Limits of quotients Find the limits in Exercises 23–42.

Exercise 23

$$\lim_{x\to 5} \frac{x - 5}{x^2 - 25}$$

Exercise 28

$$\lim_{t\to -1} \frac{t^2 + 3t + 2}{t^2 - t - 2}$$

Exercise 32

$$\lim_{x\to 0} \frac{\frac{1}{x - 1} + \frac{1}{x + 1}}{x}$$

Exercise 40

$$\lim_{x\to -2} \frac{x + 2}{\sqrt{x^2 + 5} - 3}$$

Directions for Exercises 43–50.
Limits with trigonometric functions Find the limits in Exercises 43–50.

Exercise 44

$$\lim_{x \to 0} \sin^2 x$$

Exercise 45

$$\lim_{x \to 0} \sec x$$

Exercise 54

Suppose $\lim_{x \to 4} f(x) = 0$ and $\lim_{x \to 4} g(x) = -3$. Find

a. $\lim_{x \to 4} (g(x) + 3)$

b. $\lim_{x \to 4} x f(x)$

c. $\lim_{x \to 4} (g(x))^2$

d. $\lim_{x \to 4} \dfrac{g(x)}{f(x) - 1}$

Exercise 57: Limits of Average Rates of Change

Directions for Exercises 57–62.
Because of their connection with secant lines, tangents, and instantaneous rates, limits of the form

$$\lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$$

occur frequently in calculus. In Exercises 57–62, evaluate this limit for the given value of x and function f.

Exercise 57

$$f(x) = x^2, \quad x = 1$$

Exercise 65 • (a) only

a. It can be shown that the inequalities

$$1 - \frac{x^2}{6} < \frac{x \sin x}{2 - 2 \cos x} < 1$$

hold for all values of x close to zero. What, if anything, does this tell you about

$$\lim_{x \to 0} \frac{x \sin x}{2 - 2 \cos x}?$$

Give reasons for your answer.

Exercises 75, 77, 80: Theory and Examples

Exercise 75

If $x^4 \le f(x) \le x^2$ for x in $[-1, 1]$ and $x^2 \le f(x) \le x^4$ for $x < -1$ and $x > 1$, at what points c do you automatically know $\lim_{x \to c} f(x)$? What can you say about the value of the limit at these points?

Exercise 77

If $\lim_{x \to 4} \dfrac{f(x) - 5}{x - 2} = 1$, find $\lim_{x \to 4} f(x)$.

Exercise 80

If $\lim_{x \to 0} \dfrac{f(x)}{x^2} = 1$, find

a. $\lim_{x \to 0} f(x)$

b. $\lim_{x \to 0} \dfrac{f(x)}{x}$

2.4 One-Sided Limits

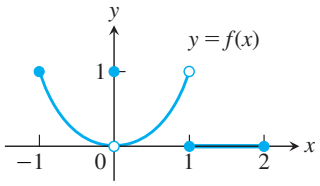
Assigned: 1, 5, 6, 17, 22, 30, 34, 36, 48–50

Thomas' Calculus, 14th SI edition • Section 2.4 • textbook pp. 88–90

Exercises 1, 5, 6: Finding Limits Graphically

Exercise 1

Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?



a. $\lim_{x \to -1^+} f(x) = 1$

b. $\lim_{x \to 0^-} f(x) = 0$

c. $\lim_{x \to 0^-} f(x) = 1$

d. $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x)$

e. $\lim_{x \to 0} f(x)$ exists.

f. $\lim_{x \to 0} f(x) = 0$

g. $\lim_{x \to 0} f(x) = 1$

h. $\lim_{x \to 1} f(x) = 1$

i. $\lim_{x \to 1} f(x) = 0$

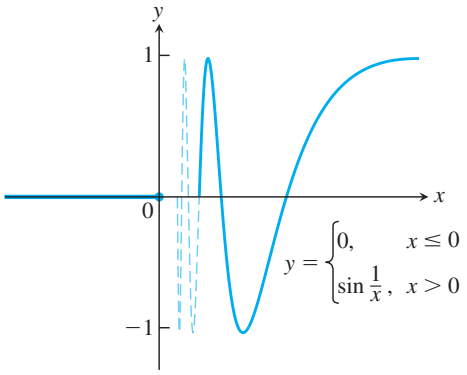
j. $\lim_{x \to 2^-} f(x) = 2$

k. $\lim_{x \to -1^-} f(x)$ does not exist.

l. $\lim_{x \to 2^+} f(x) = 0$

Exercise 5

Let $f(x) = \begin{cases} 0, & x \le 0 \\ \sin \dfrac{1}{x}, & x > 0. \end{cases}$



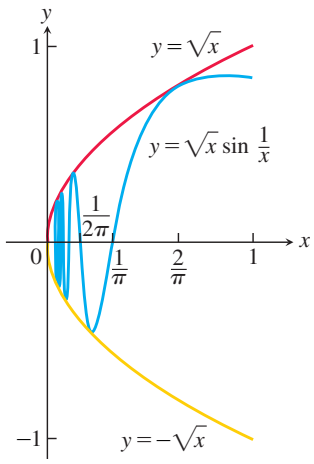
a. Does $\lim_{x \to 0^+} f(x)$ exist? If so, what is it? If not, why not?

b. Does $\lim_{x \to 0^-} f(x)$ exist? If so, what is it? If not, why not?

c. Does $\lim_{x \to 0} f(x)$ exist? If so, what is it? If not, why not?

Exercise 6

Let $g(x) = \sqrt{x} \sin(1/x)$.



a. Does $\lim_{x \to 0^+} g(x)$ exist? If so, what is it? If not, why not?

b. Does $\lim_{x \to 0^-} g(x)$ exist? If so, what is it? If not, why not?

c. Does $\lim_{x \to 0} g(x)$ exist? If so, what is it? If not, why not?

Exercises 17, 22, 30, 34, 36: Finding One-Sided Limits Algebraically

Directions for Exercises 11–20.
Find the limits in Exercises 11–20.

Exercise 17

a. $\lim_{x \to -2^+} (x + 3) \dfrac{|x + 2|}{x + 2}$

b. $\lim_{x \to -2^-} (x + 3) \dfrac{|x + 2|}{x + 2}$

Directions for Exercises 21–22.
Use the graph of the greatest integer function $y = \lfloor x \rfloor$, Figure 1.10 in Section 1.1, to help you find the limits in Exercises 21 and 22.

Exercise 22

a. $\lim_{t \to 4^+} (t - \lfloor t \rfloor)$

b. $\lim_{t \to 4^-} (t - \lfloor t \rfloor)$

Directions for Exercises 23–46.
Find the limits in Exercises 23–46.

Exercise 30

$$\lim_{x \to 0} 6x^2(\cot x)(\csc 2x)$$

Exercise 34

$$\lim_{x \to 0} \frac{x - x\cos x}{\sin^2 3x}$$

Exercise 36

$$\lim_{h \to 0} \frac{\sin(\sin h)}{\sin h}$$

Exercises 48–50: Theory and Examples

Exercise 48

If you know that $\lim_{x \to c} f(x)$ exists, can you find its value by calculating $\lim_{x \to c^+} f(x)$? Give reasons for your answer.

Exercise 49

Suppose that f is an odd function of x. Does knowing that $\lim_{x \to 0^+} f(x) = 3$ tell you anything about $\lim_{x \to 0^-} f(x)$? Give reasons for your answer.

Exercise 50

Suppose that f is an even function of x. Does knowing that $\lim_{x \to 2^-} f(x) = 7$ tell you anything about either $\lim_{x \to -2^-} f(x)$ or $\lim_{x \to -2^+} f(x)$? Give reasons for your answer.

2.5 Continuity

Assigned: 7–10, 26, 27, 32, 41, 45, 49, 59, 61, 64

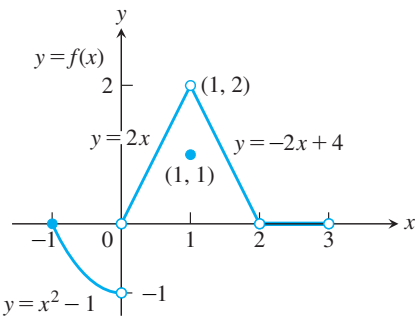
Thomas' Calculus, 14th SI edition • Section 2.5 • textbook pp. 99–101

Exercises 7–10: Continuity from Graphs

Directions for Exercises 5–10.
Exercises 5–10 refer to the function

$$\mathbf{f}(x) = \begin{cases} x^2 - 1, & -1 \le x < 0 \\ 2x, & 0 < x < 1 \\ 1, & x = 1 \\ -2x + 4, & 1 < x < 2 \\ 0, & 2 < x < 3 \end{cases}$$

graphed in the accompanying figure.



The graph for Exercises 5–10.

Exercise 7

a. Is $\mathbf{f}$ defined at $x = 2$? (Look at the definition of $\mathbf{f}$.)

b. Is $\mathbf{f}$ continuous at $x = 2$?

Exercise 8

At what values of x is $\mathbf{f}$ continuous?

Exercise 9

What value should be assigned to $\mathbf{f}(2)$ to make the extended function continuous at $x = 2$?

Exercise 10

To what new value should $\mathbf{f}(1)$ be changed to remove the discontinuity?

Exercises 26, 27: Applying the Continuity Test

Directions for Exercises 13–30.
At what points are the functions in Exercises 13–30 continuous?

Exercise 26

$$y = \sqrt[4]{3x - 1}$$

Exercise 27

$$y = (2x - 1)^{1/3}$$

Exercise 32: Limits Involving Trigonometric Functions

Directions for Exercises 31–36.
Find the limits in Exercises 31–36. Are the functions continuous at the point being approached?

Exercise 32

$$\lim_{t \to 0} \sin\left(\frac{\pi}{2}\cos(\tan t)\right)$$

Exercises 41, 45: Continuous Extensions

Exercise 41

For what value of a is

$$f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \ge 3 \end{cases}$$

continuous at every x?

Exercise 45

For what values of a and b is

$$f(x) = \begin{cases} -2, & x \le -1 \\ ax - b, & -1 < x < 1 \\ 3, & x \ge 1 \end{cases}$$

continuous at every x?

Exercises 49, 59, 61, 64: Theory and Examples

Exercise 49

Roots of a cubic Show that the equation $x^3 - 15x + 1 = 0$ has three solutions in the interval $[-4, 4]$.

Exercise 59

Never-zero continuous functions Is it true that a continuous function that is never zero on an interval never changes sign on that interval? Give reasons for your answer.

Exercise 61

A fixed point theorem Suppose that a function f is continuous on the closed interval $[0, 1]$ and that $0 \le f(x) \le 1$ for every x in $[0, 1]$. Show that there must exist a number c in $[0, 1]$ such that $f(c) = c$ (c is called a **fixed point** of f).

Reference context: Exercise 63.

Prove that f is continuous at c if and only if

$$\lim_{h \to 0} f(c + h) = f(c).$$

Exercise 64

Use Exercise 63 together with the identities

$$\sin(h + c) = \sin h \cos c + \cos h \sin c,$$
$$\cos(h + c) = \cos h \cos c - \sin h \sin c$$

to prove that both $f(x) = \sin x$ and $g(x) = \cos x$ are continuous at every point $x = c$.

2.6 Limits Involving Infinity; Asymptotes of Graphs

Assigned: 2, 10, 11, 16, 21, 23, 31, 36, 39, 43, 63, 67, 73, 74, 84, 86, 103

Thomas' Calculus, 14th SI edition • Section 2.6 • textbook pp. 111–114

Exercises 2, 10, 11: Finding Limits

Exercise 2

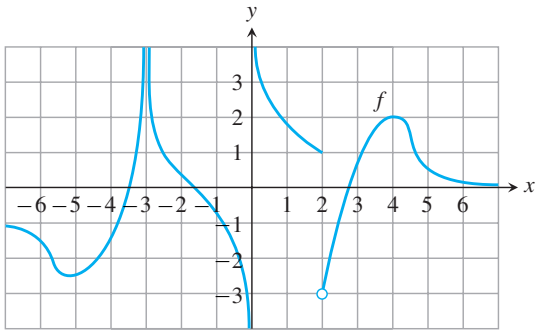
For the function $\mathbf{f}$ whose graph is given, determine the following limits.

a. $\lim_{x\to 4} \mathbf{f}(x)$ **b.** $\lim_{x\to 2^+} \mathbf{f}(x)$ **c.** $\lim_{x\to 2^-} \mathbf{f}(x)$

d. $\lim_{x\to 2} \mathbf{f}(x)$ **e.** $\lim_{x\to -3^+} \mathbf{f}(x)$ **f.** $\lim_{x\to -3^-} \mathbf{f}(x)$

g. $\lim_{x\to -3} \mathbf{f}(x)$ **h.** $\lim_{x\to 0^+} \mathbf{f}(x)$ **i.** $\lim_{x\to 0^-} \mathbf{f}(x)$

j. $\lim_{x\to 0} \mathbf{f}(x)$ **k.** $\lim_{x\to \infty} \mathbf{f}(x)$ **l.** $\lim_{x\to -\infty} \mathbf{f}(x)$



Directions for Exercises 9–12.
Find the limits in Exercises 9–12.

Exercise 10

$$\lim_{\theta\to -\infty} \frac{\cos\theta}{3\theta}$$

Exercise 11

$$\lim_{t\to -\infty} \frac{2 - t + \sin t}{t + \cos t}$$

Exercises 16, 21: Limits of Rational Functions

Directions for Exercises 13–22.
In Exercises 13–22, find the limit of each rational function **(a)** as $x \to \infty$ and **(b)** as $x \to -\infty$.

Exercise 16

$$\mathbf{f}(x) = \frac{3x + 7}{x^2 - 2}$$

Exercise 21

$$f(x) = \frac{3x^7 + 5x^2 - 1}{6x^3 - 7x + 3}$$

Exercises 23, 31, 36: Limits as or

Directions for Exercises 23–36.
The process by which we determine limits of rational functions applies equally well to ratios containing noninteger or negative powers of x: Divide numerator and denominator by the highest power of x in the denominator and proceed from there. Find the limits in Exercises 23–36.

Exercise 23

$$\lim_{x\to \infty} \sqrt{\frac{8x^2 - 3}{2x^2 + x}}$$

Exercise 31

$$\lim_{x\to \infty} \frac{2x^{5/3} - x^{1/3} + 7}{x^{8/5} + 3x + \sqrt{x}}$$

Exercise 36

$$\lim_{x\to -\infty} \frac{4 - 3x^3}{\sqrt{x^6 + 9}}$$

Exercises 39, 43: Infinite Limits

Directions for Exercises 37–48.
Find the limits in Exercises 37–48.

Exercise 39

$$\lim_{x \to 2^-} \frac{3}{x - 2}$$

Exercise 43

$$\lim_{x \to 7} \frac{4}{(x - 7)^2}$$

Exercises 63, 67: Graphing Simple Rational Functions

Directions for Exercises 63–68.
Graph the rational functions in Exercises 63–68. Include the graphs and equations of the asymptotes and dominant terms.

Exercise 63

$$y = \frac{1}{x - 1}$$

Exercise 67

$$y = \frac{x + 3}{x + 2}$$

Exercises 73, 74: Inventing Graphs and Functions

Directions for Exercises 73–76.
In Exercises 73–76, sketch the graph of a function $y = \mathbf{f}(x)$ that satisfies the given conditions. No formulas are required—just label the coordinate axes and sketch an appropriate graph. (The answers are not unique, so your graphs may not be exactly like those in the answer section.)

Exercise 73

$\mathbf{f}(0) = 0$, $\mathbf{f}(1) = 2$, $\mathbf{f}(-1) = -2$, $\lim_{x \to -\infty} \mathbf{f}(x) = -1$, and $\lim_{x \to \infty} \mathbf{f}(x) = 1$

Exercise 74

$\mathbf{f}(0) = 0$, $\lim_{x \to \pm\infty} \mathbf{f}(x) = 0$, $\lim_{x \to 0^+} \mathbf{f}(x) = 2$, and $\lim_{x \to 0^-} \mathbf{f}(x) = -2$

Exercises 84, 86: Finding Limits of Differences When

Directions for Exercises 84–90.
Find the limits in Exercises 84–90. (*Hint*: Try multiplying and dividing by the conjugate.)

Exercise 84

$$\lim_{x \to \infty} \left(\sqrt{x + 9} - \sqrt{x + 4}\right)$$

Exercise 86

$$\lim_{x \to -\infty} \left(\sqrt{x^2 + 3} + x\right)$$

Exercise 103: Oblique Asymptotes

Directions for Exercises 103–108.
Graph the rational functions in Exercises 103–108. Include the graphs and equations of the asymptotes.

Exercise 103

$$y = \frac{x^2}{x - 1}$$

3.1 Tangent Lines and the Derivative at a Point

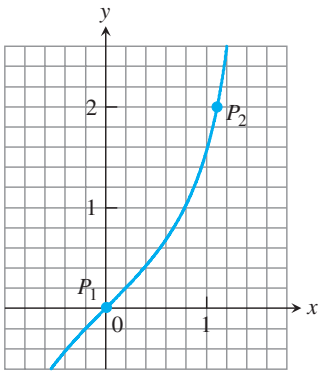
Assigned: 1, 2, 23, 33, 41, 47

Thomas' Calculus, 14th SI edition • Section 3.1 • textbook pp. 122–124

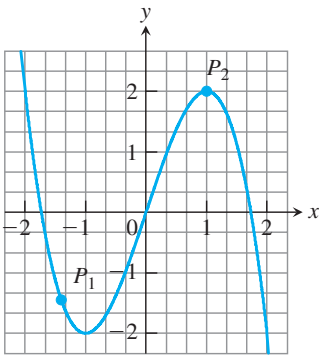
Exercises 1, 2: Slopes and Tangent Lines

Directions for Exercises 1–4.
In Exercises 1–4, use the grid and a straight edge to make a rough estimate of the slope of the curve (in y-units per x-unit) at the points P_1 and P_2.

Exercise 1

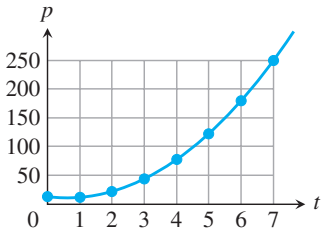


Exercise 2



Exercise 23

Growth of yeast cells In a controlled laboratory experiment, yeast cells are grown in an automated cell culture system that counts the number P of cells present at hourly intervals. The number after t hours is shown in the accompanying figure.



a. Explain what is meant by the derivative $P'(5)$. What are its units?

b. Which is larger, $P'(2)$ or $P'(3)$? Give a reason for your answer.

c. The quadratic curve capturing the trend of the data points (see Section 1.4) is given by $P(t) = 6.10t^2 - 9.28t + 16.43$. Find the instantaneous rate of growth when $t = 5$ hours.

Exercise 33

Show that the line $y = mx + b$ is its own tangent line at any point $(x_0, mx_0 + b)$.

Exercises 41, 47: Vertical Tangent Lines

Directions for Exercises 39–48.
Graph the curves in Exercises 39–48.

a. Where do the graphs appear to have vertical tangent lines?

Exercise 41

$$y = x^{1/5}$$

Exercise 47

$$y = \begin{cases} -\sqrt{|x|}, & x \le 0 \\ \sqrt{x}, & x > 0 \end{cases}$$

3.2 The Derivative as a Function

Assigned: 1, 20, 27–31, 34, 45, 46, 48–50, 59, 60

Thomas' Calculus, 14th SI edition • Section 3.2 • textbook pp. 129–133

Exercise 1: Finding Derivative Functions and Values

Directions for Exercises 1–6.
Exercises 1–6. Then find the values of the derivatives as specified.

Exercise 1

$f(x) = 4 - x^2$; $f'(-3)$, $f'(0)$, $f'(1)$

Exercise 20: Slopes and Tangent Lines

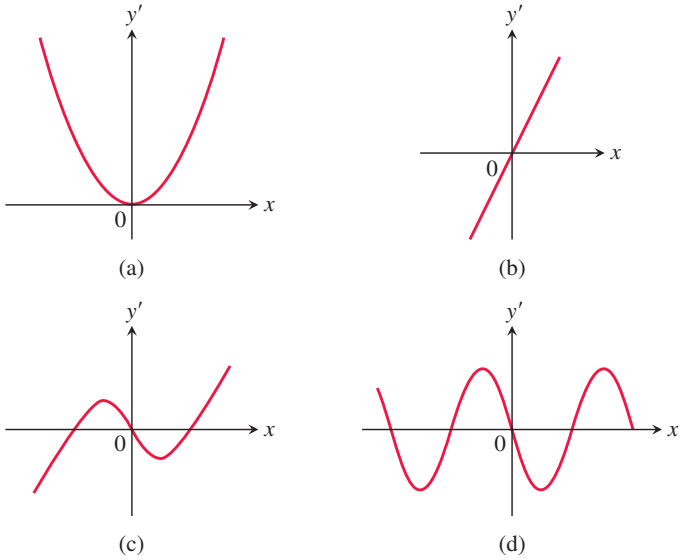
Directions for Exercises 19–22.
In Exercises 19–22, find the values of the derivatives.

Exercise 20

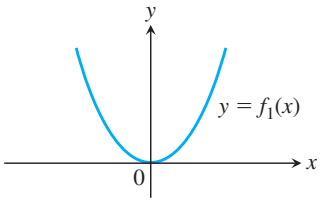
$$\left.\frac{dy}{dx}\right|_{x=\sqrt{3}} \quad \text{if} \quad y = 1 - \frac{1}{x}$$

Exercises 27–31, 34: Graphs

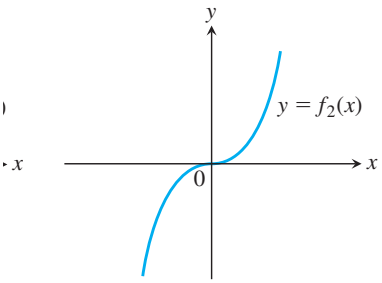
Directions for Exercises 27–30.
Match the functions graphed in Exercises 27–30 with the derivatives graphed in the accompanying figures (a)–(d).



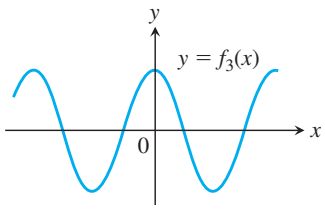
Exercise 27



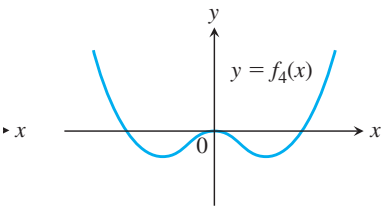
Exercise 28



Exercise 29

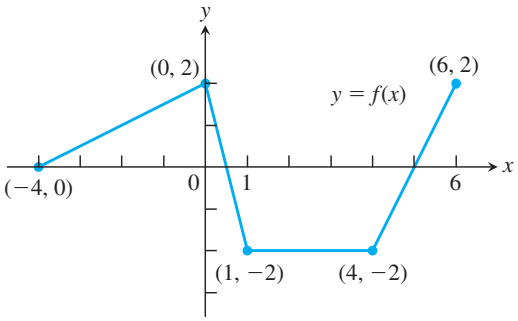


Exercise 30



Exercise 31

a. The graph in the accompanying figure is made of line segments joined end to end. At which points of the interval $[-4, 6]$ is f' not defined? Give reasons for your answer.

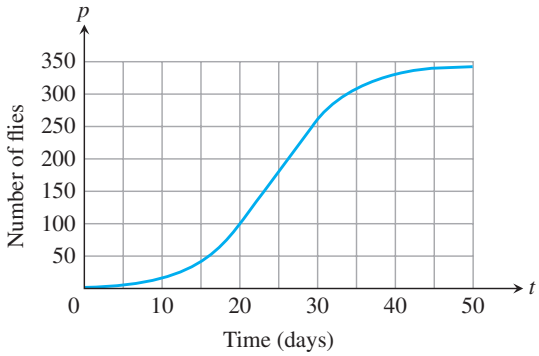


b. Graph the derivative of **f**.
The graph should show a step function.

Exercise 34

Fruit flies (*Continuation of Example 4, Section 2.1.*) Populations starting out in closed environments grow slowly at first, when there are relatively few members, then more rapidly as the number of reproducing individuals increases and resources are still abundant, then slowly again as the population reaches the carrying capacity of the environment.

a. Use the graphical technique of Example 3 to graph the derivative of the fruit fly population. The graph of the population is reproduced here.



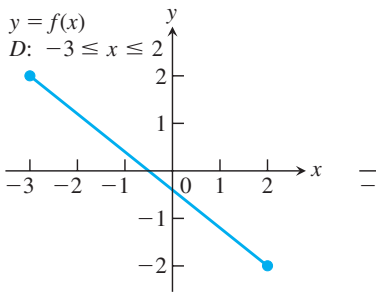
b. During what days does the population seem to be increasing fastest? Slowest?

Exercises 45, 46, 48–50: Differentiability and Continuity on an Interval

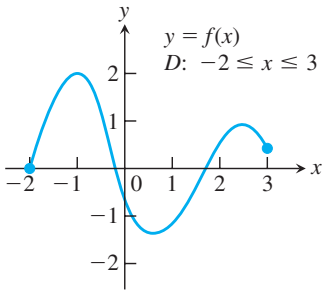
Directions for Exercises 45–50.
Each figure in Exercises 45–50 shows the graph of a function over a closed interval D. At what domain points does the function appear to be

- **a.** differentiable?
- **b.** continuous but not differentiable?
- **c.** neither continuous nor differentiable?

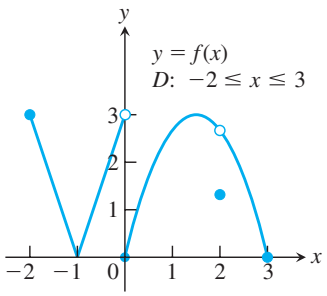
Exercise 45



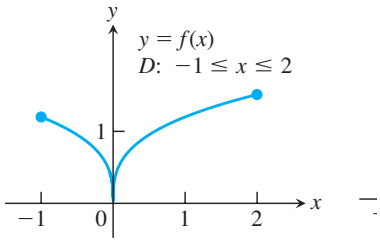
Exercise 46



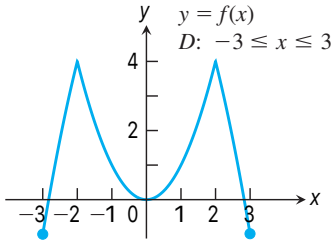
Exercise 48



Exercise 49



Exercise 50



Exercises 59, 60: Theory and Examples

Exercise 59

Limit of a quotient Suppose that functions $g(t)$ and $h(t)$ are defined for all values of t and $g(0) = h(0) = 0$. Can $\lim_{t\to 0} (g(t))/(h(t))$ exist? If it does exist, must it equal zero? Give reasons for your answers.

Exercise 60

- **a.** Let $f(x)$ be a function satisfying $|f(x)| \le x^2$ for $-1 \le x \le 1$. Show that f is differentiable at $x = 0$ and find $f'(0)$.
- **b.** Show that

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is differentiable at $x = 0$ and find $f'(0)$.

3.3 Differentiation Rules

Assigned: 6, 11, 17, 20, 21, 29, 35, 39, 41, 46, 55–58

Thomas' Calculus, 14th SI edition • Section 3.3 • textbook pp. 139–142

Exercises 6, 11, 17, 20, 21, 29, 35, 39: Derivative Calculations

Directions for Exercises 1–12.
In Exercises 1–12, find the first and second derivatives.

Exercise 6

$$y = \frac{x^3}{3} + \frac{x^2}{2} + \frac{x}{4}$$

Exercise 11

$$r = \frac{1}{3s^2} - \frac{5}{2s}$$

Directions for Exercises 17–40.
Find the derivatives of the functions in Exercises 17–40.

Exercise 17

$$y = \frac{2x + 5}{3x - 2}$$

Exercise 20

$$f(t) = \frac{t^2 - 1}{t^2 + t - 2}$$

Exercise 21

$$v = (1 - t)\,(1 + t^2)^{-1}$$

Directions for Exercises 29–32.
Find the derivatives of all orders of the functions in Exercises 29–32.

Exercise 29

$$y = \frac{x^4}{2} - \frac{3}{2}x^2 - x$$

Directions for Exercises 33–38.
Find the first and second derivatives of the functions in Exercises 33–38.

Exercise 35

$$r = \frac{(\theta - 1)(\theta^2 + \theta + 1)}{\theta^3}$$

Directions for Exercises 17–40.
Find the derivatives of the functions in Exercises 17–40.

Exercise 39

Suppose u and v are functions of x that are differentiable at $x = 0$ and that

$$u(0) = 5, \quad u'(0) = -3, \quad v(0) = -1, \quad v'(0) = 2.$$

Find the values of the following derivatives at $x = 0$.

a. $\frac{d}{dx}(uv)$ **b.** $\frac{d}{dx}\left(\frac{u}{v}\right)$ **c.** $\frac{d}{dx}\left(\frac{v}{u}\right)$ **d.** $\frac{d}{dx}(7v - 2u)$

Exercises 41, 46: Slopes and Tangent Lines

Exercise 41

a. Normal line to a curve Find an equation for the line perpendicular to the tangent line to the curve $y = x^3 - 4x + 1$ at the point $(2, 1)$.

b. Smallest slope What is the smallest slope on the curve? At what point on the curve does the curve have this slope?

c. Tangent lines having specified slope Find equations for the tangent lines to the curve at the points where the slope of the curve is 8.

Exercise 46

Quadratics having a common tangent The curves $y = x^2 + ax + b$ and $y = cx - x^2$ have a common tangent line at the point $(1, 0)$. Find a, b, and c.

Exercises 55–58: Theory and Examples

Directions for Exercises 55–56.
For Exercises 55 and 56 evaluate each limit by first converting each to a derivative at a particular x-value.

Exercise 55

$$\lim_{x \to 1} \frac{x^{50} - 1}{x - 1}$$

Exercise 56

$$\lim_{x \to -1} \frac{x^{2/9} - 1}{x + 1}$$

Exercise 57

Find the value of a that makes the following function differentiable for all x-values.

$$g(x) = \begin{cases} ax, & \text{if } x < 0 \\ x^2 - 3x, & \text{if } x \geq 0 \end{cases}$$

Exercise 58

Find the values of a and b that make the following function differentiable for all x-values.

$$f(x) = \begin{cases} ax + b, & x > -1 \\ bx^2 - 3, & x \leq -1 \end{cases}$$

3.4 The Derivative as a Rate of Change

Assigned: 3, 7, 15, 21, 23, 25, 26

Thomas' Calculus, 14th SI edition • Section 3.4 • textbook pp. 148–152

Exercises 3, 7: Motion Along a Coordinate Line

Directions for Exercises 1–6.
Exercises 1–6 give the positions $s = f(t)$ of a body moving on a coordinate line, with s in meters and t in seconds.

a. Find the body's displacement and average velocity for the given time interval.

b. Find the body's speed and acceleration at the endpoints of the interval.

c. When, if ever, during the interval does the body change direction?

Exercise 3

$s = -t^3 + 3t^2 - 3t, \quad 0 \le t \le 3$

Exercise 7

Particle motion At time t, the position of a body moving along the s-axis is $s = t^3 - 6t^2 + 9t$ m.

a. Find the body's acceleration each time the velocity is zero.

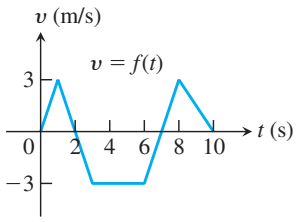
b. Find the body's speed each time the acceleration is zero.

c. Find the total distance traveled by the body from $t = 0$ to $t = 2$.

Exercises 15, 21: Understanding Motion from Graphs

Exercise 15

The accompanying figure shows the velocity $v = ds/dt = f(t)$ (m/s) of a body moving along a coordinate line.



a. When does the body reverse direction?

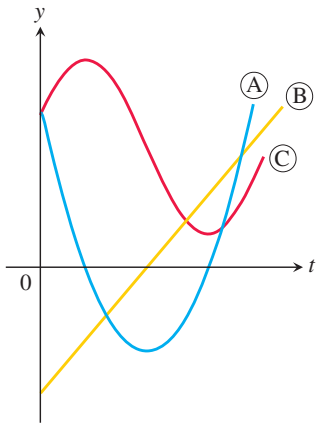
b. When (approximately) is the body moving at a constant speed?

c. Graph the body's speed for $0 \le t \le 10$.

d. Graph the acceleration, where defined.

Exercise 21

The graphs in the accompanying figure show the position s, velocity $v = ds/dt$, and acceleration $a = d^2s/dt^2$ of a body moving along a coordinate line as functions of time t. Which graph is which? Give reasons for your answers.



Exercise 23

Marginal cost Suppose that the dollar cost of producing x washing machines is $c(x) = 2000 + 100x - 0.1x^2$.

a. Find the average cost per machine of producing the first 100 washing machines.

b. Find the marginal cost when 100 washing machines are produced.

c. Show that the marginal cost when 100 washing machines are produced is approximately the cost of producing one more washing machine after the first 100 have been made, by calculating the latter cost directly.

Exercises 25, 26: Additional Applications

Exercise 25

Bacterium population When a bactericide was added to a nutrient broth in which bacteria were growing, the bacterium population continued to grow for a while, but then stopped growing and began to decline. The size of the population at time t (hours) was $b = 10^6 + 10^4 t - 10^3 t^2$. Find the growth rates at

a. $t = 0$ hours.

b. $t = 5$ hours.

c. $t = 10$ hours.

Exercise 26

Body surface area A typical male's body surface area S in square meters is often modeled by the formula $S = \frac{1}{60}\sqrt{wh}$, where h is the height in cm, and w the weight in kg, of the person. Find the rate of change of body surface area with respect to weight

for males of constant height $h = 180$ cm. Does S increase more rapidly with respect to weight at lower or higher body weights? Explain.

3.5 Derivatives of Trigonometric Functions

Assigned: 3, 12, 15, 17, 26, 32, 34, 47, 52, 53, 58, 60, 61 • Specified subparts are marked below.

Thomas' Calculus, 14th SI edition • Section 3.5 • textbook pp. 156–158

Exercises 3, 12, 15, 17, 26, 32, 34: Derivatives

Directions for Exercises 1–18.
In Exercises 1–18, find dy/dx.

Exercise 3

$$y = x^2 \cos x$$

Exercise 12

$$y = \frac{\cos x}{1 + \sin x}$$

Exercise 15

$$y = (\sec x + \tan x)(\sec x - \tan x)$$

Exercise 17

$$f(x) = x^3 \sin x \cos x$$

Directions for Exercises 23–26.
In Exercises 23–26, find $dr/d\theta$.

Exercise 26

$$r = (1 + \sec\theta)\sin\theta$$

Directions for Exercises 27–32.
In Exercises 27–32, find dp/dq.

Exercise 32

$$p = \frac{3q + \tan q}{q \sec q}$$

Exercise 34

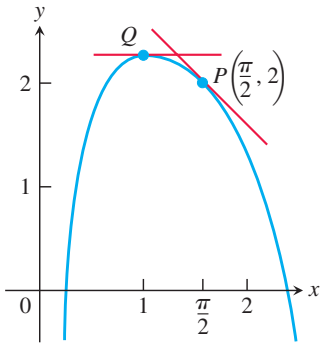
Find $y^{(4)} = d^4y/dx^4$ if

a. $y = -2\sin x$. **b.** $y = 9\cos x$.

Exercise 47: Tangent Lines

Directions for Exercises 47–48.
In Exercises 47 and 48, find an equation for **(a)** the tangent to the curve at P and **(b)** the horizontal tangent to the curve at Q.

Exercise 47



$y = 4 + \cot x - 2\csc x$

Exercises 52, 53: Trigonometric Limits

Directions for Exercises 49–56.
Find the limits in Exercises 49–56.

Exercise 52

$$\lim_{\theta \to \pi/4} \frac{\tan\theta - 1}{\theta - \frac{\pi}{4}}$$

Exercise 53

$$\lim_{x \to 0} \sec\left[\cos x + \pi \tan\left(\frac{\pi}{4 \sec x}\right) - 1\right]$$

Exercises 58, 60, 61: Theory and Examples

Directions for Exercises 57–58.
The equations in Exercises 57 and 58 give the position $s = f(t)$ of a body moving on a coordinate line (s in meters, t in seconds). Find the body's velocity, speed, acceleration, and jerk at time $t = \pi/4$ s.

Exercise 58

$$s = \sin t + \cos t$$

Exercise 60

Is there a value of b that will make

$$g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \geq 0 \end{cases}$$

continuous at $x = 0$? Differentiable at $x = 0$? Give reasons for your answers.

Exercise 61 • (a) only

By computing the first few derivatives and looking for a pattern, find the following derivatives.

a. $\dfrac{d^{999}}{dx^{999}}(\cos x)$

3.6 The Chain Rule

Assigned: 5, 7, 10, 15, 29, 34, 41, 54, 67, 72, 75, 76, 84, 86

Thomas' Calculus, 14th SI edition • Section 3.6 • textbook pp. 163–166

Exercises 5, 7, 10, 15, 29, 34, 41, 54: Derivative Calculations

Directions for Exercises 1–8.
In Exercises 1–8, given $y = f(u)$ and $u = g(x)$, find $dy/dx = dy/dx = f'(g(x))g'(x)$.

Exercise 5

$$y = \sqrt{u}, \quad u = \sin x$$

Exercise 7

$$y = \tan u, \quad u = \pi x^2$$

Directions for Exercises 9–18.
In Exercises 9–18, write the function in the form $y = f(u)$ and $u = g(x)$. Then find dy/dx as a function of x.

Exercise 10

$$y = (4 - 3x)^9$$

Exercise 15

$$y = \sec(\tan x)$$

Directions for Exercises 19–40.
Find the derivatives of the functions in Exercises 19–40.

Exercise 29

$$y = (4x + 3)^4(x + 1)^{-3}$$

Exercise 34

$$g(x) = \frac{\tan 3x}{(x + 7)^4}$$

Directions for Exercises 41–58.
In Exercises 41–58, find dy/dt.

Exercise 41

$$y = \sin^2(\pi t - 2)$$

Exercise 54

$$y = 4\sin\left(\sqrt{1 + \sqrt{t}}\right)$$

Exercises 67, 72, 75, 76: Finding Derivative Values

Directions for Exercises 67–72.
In Exercises 67–72, find the value of $(f \circ g)'$ at the given value of x.

Exercise 67

$$f(u) = u^5 + 1, \quad u = g(x) = \sqrt{x}, \quad x = 1$$

Exercise 72

$$f(u) = \left(\frac{u - 1}{u + 1}\right)^2, \quad u = g(x) = \frac{1}{x^2} - 1, \quad x = -1$$

Exercise 75

Suppose that functions f and g and their derivatives with respect to x have the following values at $x = 2$ and $x = 3$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	8	2	$1/3$	-3
3	3	-4	2π	5

Find the derivatives with respect to x of the following combinations at the given value of x.

a. $2f(x), \quad x = 2$
b. $f(x) + g(x), \quad x = 3$
c. $f(x) \cdot g(x), \quad x = 3$
d. $f(x)/g(x), \quad x = 2$
e. $f(g(x)), \quad x = 2$
f. $\sqrt{f(x)}, \quad x = 2$
g. $1/g^2(x), \quad x = 3$
h. $\sqrt{f^2(x) + g^2(x)}, \quad x = 2$

Exercise 76

Suppose that the functions f and g and their derivatives with respect to x have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	5	$1/3$
1	3	-4	$-1/3$	$-8/3$

Find the derivatives with respect to x of the following combinations at the given value of x.

a. $5f(x) - g(x), \quad x = 1$ **b.** $f(x)g^3(x), \quad x = 0$

c. $\dfrac{f(x)}{g(x) + 1}, \quad x = 1$ **d.** $f(g(x)), \quad x = 0$

e. $g(f(x)), \quad x = 0$ **f.** $(x^{11} + f(x))^{-2}, \quad x = 1$

g. $f(x + g(x)), \quad x = 0$

Exercises 84, 86: Theory and Examples

Exercise 84

Slopes on sine curves

a. Find equations for the tangents to the curves $y = \sin 2x$ and $y = -\sin(x/2)$ at the origin. Is there anything special about how the tangents are related? Give reasons for your answer.

b. Can anything be said about the tangents to the curves $y = \sin mx$ and $y = -\sin(x/m)$ at the origin (m a constant $\neq 0$)? Give reasons for your answer.

c. For a given m, what are the largest values the slopes of the curves $y = \sin mx$ and $y = -\sin(x/m)$ can ever have? Give reasons for your answer.

d. The function $y = \sin x$ completes one period on the interval $[0, 2\pi]$, the function $y = \sin 2x$ completes two periods, the function $y = \sin(x/2)$ completes half a period, and so on. Is there any relation between the number of periods $y = \sin mx$ completes on $[0, 2\pi]$ and the slope of the curve $y = \sin mx$ at the origin? Give reasons for your answer.

Exercise 86

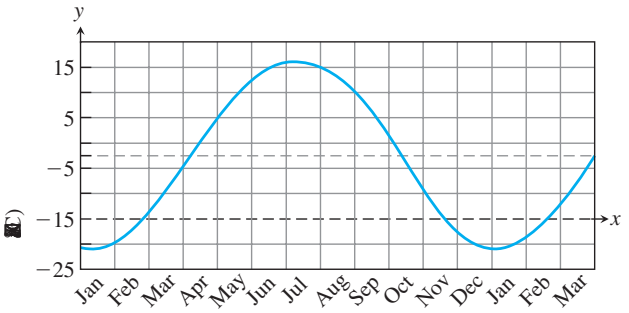
Temperatures in Fairbanks, Alaska The graph in the accompanying figure shows the average Celsius temperature in Fairbanks, Alaska, during a typical 365-day year. The equation that approximates the temperature on day x is

$$y = 20 \sin\left[\frac{2\pi}{365}(x - 101)\right] - 4$$

and is graphed in the accompanying figure.

a. On what day is the temperature increasing the fastest?

b. About how many degrees per day is the temperature increasing when it is increasing at its fastest?



3.7 Implicit Differentiation

Assigned: 1, 12, 19, 31, 43

Thomas' Calculus, 14th SI edition • Section 3.7 • textbook pp. 169–171

Exercises 1, 12: Differentiating Implicitly

Directions for Exercises 1–14.
Use implicit differentiation to find dy/dx in Exercises 1–14.

Exercise 1

$$x^2y + xy^2 = 6$$

Exercise 12

$$x^4 + \sin y = x^3y^2$$

Exercise 19: Second Derivatives

Directions for Exercises 19–26.
In Exercises 19–26, use implicit differentiation to find dy/dx and then d^2y/dx^2. Write the solutions in terms of x and y only.

Exercise 19

$$x^2 + y^2 = 1$$

Exercises 31, 43: Slopes, Tangents, and Normals

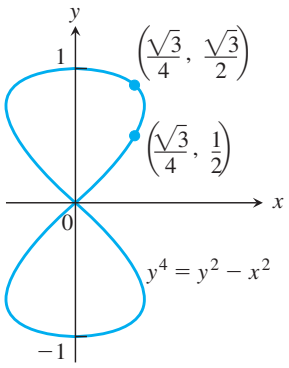
Directions for Exercises 31–40.
In Exercises 31–40, verify that the given point is on the curve and find the lines that are **(a)** tangent and **(b)** normal to the curve at the given point.

Exercise 31

$$x^2 + xy - y^2 = 1, \quad (2, 3)$$

Exercise 43

The eight curve Find the slopes of the curve $y^4 = y^2 - x^2$ at the two points shown here.



3.8 Related Rates

Assigned: 7, 23, 27, 32, 42

Thomas' Calculus, 14th SI edition • Section 3.8 • textbook pp. 176–180

Exercises 7, 23, 27, 32, 42: Related Rates

Exercise 7

If $x^2 + y^2 = 25$ and $dx/dt = -2$, then what is dy/dt when $x = 3$ and $y = -4$?

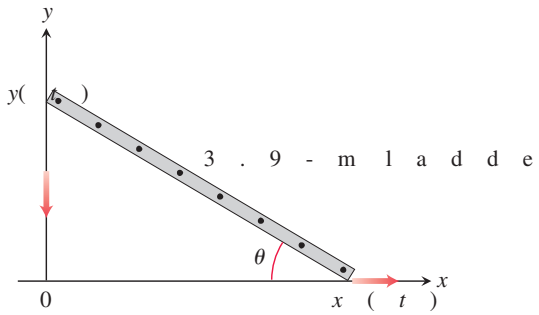
Exercise 23

A sliding ladder A 3.9-m ladder is leaning against a house when its base starts to slide away (see accompanying figure). By the time the base is 3.6 m from the house, the base is moving at the rate of 1.5 m/s.

a. How fast is the top of the ladder sliding down the wall then?

b. At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?

c. At what rate is the angle θ between the ladder and the ground changing then?



Exercise 27

A growing sand pile Sand falls from a conveyor belt at the rate of 10 m^3/min onto the top of a conical pile. The height of the pile is always three-eighths of the base diameter. How fast are the **(a)** height and **(b)** radius changing when the pile is 4 m high? Answer in centimeters per minute.

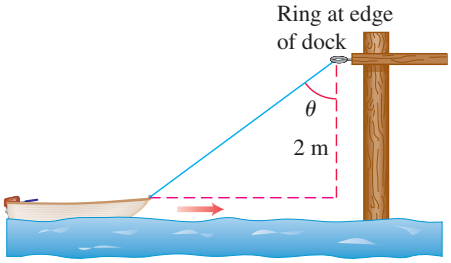
Exercise 32

Hauling in a dinghy A dinghy is pulled toward a dock by a rope from the bow through a ring on the dock 2 m above the bow. The rope is hauled in at the rate of 0.5 m/s.

a. How fast is the boat approaching the dock when 3 m of rope are out?

Exercise 32 (continued)

b. At what rate is the angle θ changing at this instant (see the figure)?



Exercise 42

Highway patrol A highway patrol plane flies 3 km above a level, straight road at a steady 120 km/h. The pilot sees an oncoming car and with radar determines that at the instant the line-of-sight distance from plane to car is 5 km, the line-of-sight distance is decreasing at the rate of 160 km/h. Find the car's speed along the highway.

3.9 Linearization and Differentials

Assigned: 1, 6, 17, 18, 29, 37, 42, 51, 56

Thomas' Calculus, 14th SI edition • Section 3.9 • textbook pp. 189–192

Exercises 1, 6: Finding Linearizations

Directions for Exercises 1–5.
In Exercises 1–5, find the linearization $L(x)$ of $f(x)$ at $x = a$.

Exercise 1

$f(x) = x^3 - 2x + 3, \quad a = 2$

Exercise 6

Common linear approximations at $x = 0$ Find the linearizations of the following functions at $x = 0$.

a. $\sin x$ **b.** $\cos x$ **c.** $\tan x$

Exercises 17, 18: Derivatives in Differential Form

Directions for Exercises 17–28.
In Exercises 17–28, find dy.

Exercise 17

$y = x^3 - 3\sqrt{x}$

Exercise 18

$y = x\sqrt{1 - x^2}$

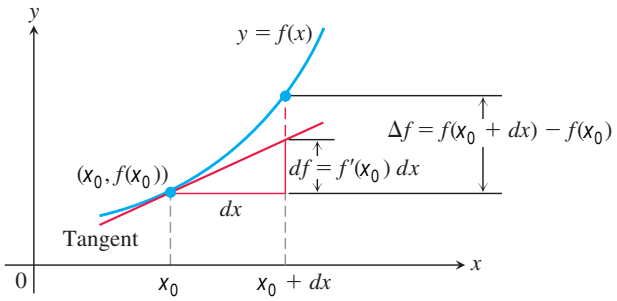
Exercise 29: Approximation Error

Directions for Exercises 29–34.
In Exercises 29–34, each function $f(x)$ changes value when x changes from x_0 to $x_0 + dx$. Find

a. the change $\Delta f = f(x_0 + dx) - f(x_0)$;

b. the value of the estimate $df = f'(x_0)\,dx$; and

c. the approximation error $|\Delta f - df|$.



Exercise 29

$f(x) = x^2 + 2x, \quad x_0 = 1, \quad dx = 0.1$

Exercise 37: Differential Estimates of Change

Directions for Exercises 35–40.
In Exercises 35–40, write a differential formula that estimates the given change in volume or surface area.

Exercise 37

The change in the surface area $S = 6x^2$ of a cube when the edge lengths change from x_0 to $x_0 + dx$

Exercises 42, 51, 56: Applications

Exercise 42

The diameter of a tree was 25 cm. During the following year, the circumference increased 5 cm. About how much did the tree's diameter increase? The tree's cross-sectional area?

Exercise 51

The effect of flight maneuvers on the heart The amount of work done by the heart's main pumping chamber, the left ventricle, is given by the equation

$$W = PV + \frac{V\delta v^2}{2g},$$

where W is the work per unit time, P is the average blood pressure, V is the volume of blood pumped out during the unit of time, δ ("delta") is the weight density of the blood, v is the average velocity of the exiting blood, and g is the acceleration of gravity.

When P, V, δ, and v remain constant, W becomes a function of g, and the equation takes the simplified form

$$W = a + \frac{b}{g} \ (a, b \text{ constant}).$$

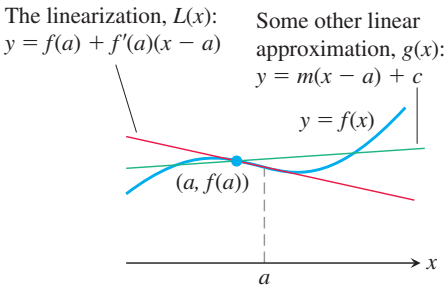
As a member of NASA's medical team, you want to know how sensitive W is to apparent changes in g caused by flight maneuvers, and this depends on the initial value of g. As part of your investigation, you decide to compare the effect on W of a given change dg on the moon, where $g = 1.6\ \text{m/s}^2$, with the effect the same change dg would have on Earth, where $g = 9.8\ \text{m/s}^2$. Use the simplified equation above to find the ratio of dW_{moon} to dW_{Earth}.

Exercise 56

The linearization is the best linear approximation Suppose that $y = f(x)$ is differentiable at $x = a$ and that $g(x) = g(x) = m(x - a) + c$ is a linear function in which m and c are constants. If the error $E(x) = f(x) - g(x)$ were small enough near $x = a$, we might think of using g as a linear approximation of f instead of the linearization $L(x) = f(a) + f'(a)(x - a)$. Show that if we impose on g the conditions

1. $E(a) = 0$ The approximation error is zero at $x = a$.
2. $\lim_{x \to a} \frac{E(x)}{x - a} = 0$ The error is negligible when compared with $x - a$.

then $g(x) = f(a) + f'(a)(x - a)$. Thus, the linearization $L(x)$ gives the only linear approximation whose error is both zero at $x = a$ and negligible in comparison with $x - a$.



4.1 Extreme Values of Functions on Closed Intervals

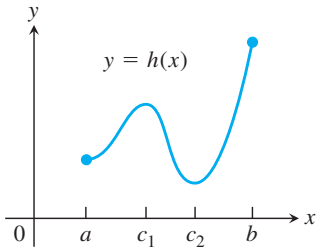
Assigned: 1–6, 20, 36, 42, 45, 59, 60

Thomas' Calculus, 14th SI edition • Section 4.1 • textbook pp. 206–209

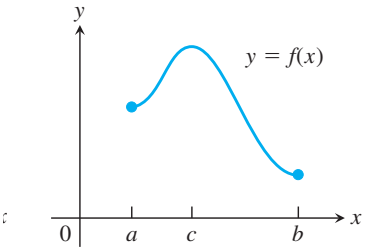
Exercises 1–6, 20: Finding Extrema from Graphs

Directions for Exercises 1–6.
In Exercises 1–6, determine from the graph whether the function has any absolute extreme values on $[a, b]$. Then explain how your answer is consistent with Theorem 1.

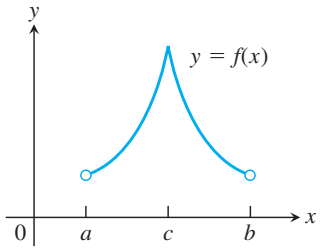
Exercise 1



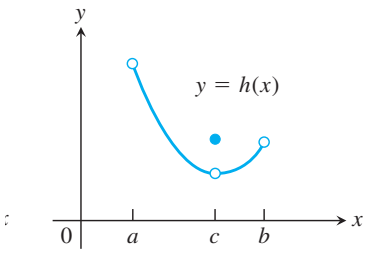
Exercise 2



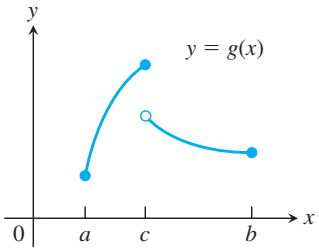
Exercise 3



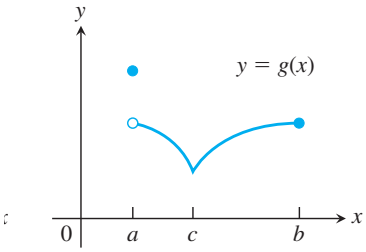
Exercise 4



Exercise 5



Exercise 6



Directions for Exercises 15–20.
In Exercises 15–20, sketch the graph of each function and determine whether the function has any absolute extreme values on its domain. Explain how your answer is consistent with Theorem 1.

Exercise 20

$$f(x) = \begin{cases} x + 1, & -1 \le x < 0 \\ \cos x, & 0 < x \le \dfrac{\pi}{2} \end{cases}$$

Exercise 36: Absolute Extrema on Finite Closed Intervals

Directions for Exercises 21–36.
In Exercises 21–36, find the absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

Exercise 36

$f(t) = |t - 5|, \quad 4 \le t \le 7$

Exercises 42, 45: Finding Critical Points

Directions for Exercises 41–50.
In Exercises 41–50, determine critical points and domain endpoints for each function.

Exercise 42

$f(x) = 6x^2 - x^3$

Exercise 45

$y = x^2 + \dfrac{2}{x}$

Exercises 59, 60: Local Extrema and Critical Points

Directions for Exercises 59–60.
In Exercises 59 and 60, give reasons for your answers.

Exercise 59

Let $f(x) = (x - 2)^{2/3}$.

a. Does $f'(2)$ exist?

b. Show that the only local extreme value of **f** occurs at $x = 2$.

c. Does the result in part (b) contradict the Extreme Value Theorem?

d. Repeat parts (a) and (b) for $f(x) = (x - a)^{2/3}$, replacing 2 by a.

Exercise 60

Let $f(x) = |x^3 - 9x|$.

a. Does $f'(0)$ exist?

b. Does $f'(3)$ exist?

c. Does $f'(-3)$ exist?

d. Determine all extrema of **f**.

4.2 The Mean Value Theorem

Assigned: 13, 15, 18, 20, 23, 27, 33, 60, 61, 63, 65 • Specified subparts are marked below.

Thomas' Calculus, 14th SI edition • Section 4.2 • textbook pp. 213–215

Exercise 13

The function

$$f(x) = \begin{cases} x, & 0 \le x < 1 \\ 0, & x = 1 \end{cases}$$

is zero at $x = 0$ and $x = 1$ and differentiable on $(0, 1)$, but its derivative on $(0, 1)$ is never zero. How can this be? Doesn't Rolle's Theorem say the derivative has to be zero somewhere in $(0, 1)$? Give reasons for your answer.

Exercises 15, 18, 20, 23: Roots (Zeros)

Exercise 15 • a (iii), b only

a. Plot the zeros of each polynomial on a line together with the zeros of its first derivative.

iii) $y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$

b. Use Rolle's Theorem to prove that between every two zeros of $x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ there lies a zero of

$$nx^{n-1} + (n - 1)a_{n-1}x^{n-2} + \cdots + a_1.$$

Exercise 18

Show that a cubic polynomial can have at most three real zeros.

Directions for Exercises 19–26.
Show that the functions in Exercises 19–26 have exactly one zero in the given interval.

Exercise 20

$$f(x) = x^3 + \frac{4}{x^2} + 7, \quad (-\infty, 0)$$

Exercise 23

$$r(\theta) = \theta + \sin^2\left(\frac{\theta}{3}\right) - 8, \quad (-\infty, \infty)$$

Exercises 27, 33: Finding Functions from Derivatives

Exercise 27

Suppose that $f(-1) = 3$ and that $f'(x) = 0$ for all x. Must $f(x) = 3$ for all x? Give reasons for your answer.

Directions for Exercises 31–36.
In Exercises 31–36, find all possible functions with the given derivative.

Exercise 33

a. $y' = -\dfrac{1}{x^2}$ **b.** $y' = 1 - \dfrac{1}{x^2}$ **c.** $y' = 5 + \dfrac{1}{x^2}$

Exercises 60, 61, 63, 65: Theory and Examples

Exercise 60

Parallel tangents Assume that f and g are differentiable on $[a, b]$ and that $f(a) = g(a)$ and $f(b) = g(b)$. Show that there is at least one point between a and b where the tangents to the graphs of f and g are parallel or the same line. Illustrate with a sketch.

Exercise 61

Suppose that $f'(x) \le 1$ for $1 \le x \le 4$. Show that $f(4) - f(1) \le 3$.

Exercise 63

Show that $|\cos x - 1| \le |x|$ for all x-values. (*Hint:* Consider $f(t) = \cos t$ on $[0, x]$.)

Exercise 65

If the graphs of two differentiable functions $f(x)$ and $g(x)$ start at the same point in the plane and the functions have the same rate of change at every point, do the graphs have to be identical? Give reasons for your answer.

4.3 Monotonic Functions and the First Derivative Test

Assigned: 15–18, 23, 35, 55, 67 • Specified subparts are marked below.

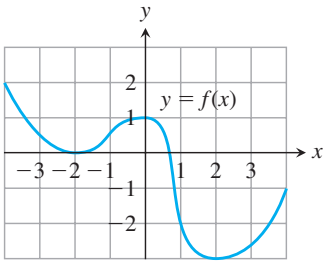
Thomas' Calculus, 14th SI edition • Section 4.3 • textbook pp. 219–220

Exercises 15–18, 23, 35, 55: Identifying Extrema

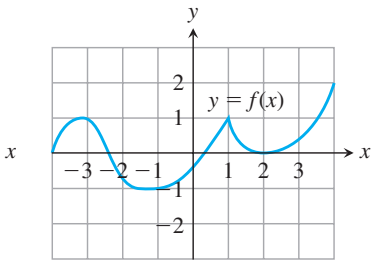
Directions for Exercises 15–40.
In Exercises 15–40:

- **a.** Find the open intervals on which the function is increasing and decreasing.
- **b.** Identify the function's local and absolute extreme values, if any, saying where they occur.

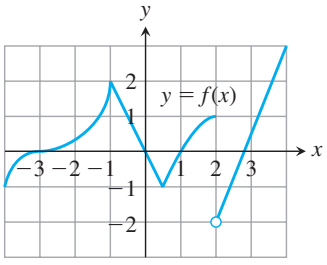
Exercise 15



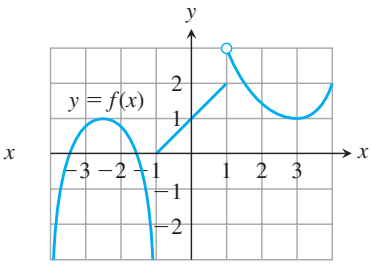
Exercise 16



Exercise 17



Exercise 18



Exercise 23

$$f(\theta) = 3\theta^2 - 4\theta^3$$

Exercise 35

$$f(x) = \frac{x^2 - 3}{x - 2}, \quad x \neq 2$$

Directions for Exercises 53–60.
In Exercises 53–60:

- **a.** Find the local extrema of each function on the given interval, and say where they occur.

Exercise 55 • (a) only

$$f(x) = \sqrt{3}\cos x + \sin x, \quad 0 \le x \le 2\pi$$

Exercise 67

Sketch the graph of a continuous function $y = g(x)$ such that

- **a.** $g(2) = 2$, $0 < g' < 1$ for $x < 2$, $g'(x) \to 1^-$ as $x \to 2^-$, $-1 < g' < 0$ for $x > 2$, and $g'(x) \to -1^+$ as $x \to 2^+$;
- **b.** $g(2) = 2$, $g' < 0$ for $x < 2$, $g'(x) \to -\infty$ as $x \to 2^-$, $g' > 0$ for $x > 2$, and $g'(x) \to \infty$ as $x \to 2^+$.

4.4 Concavity and Curve Sketching

Assigned: 1, 4, 17, 23, 32, 46, 75, 78, 95, 97, 106, 107, 112

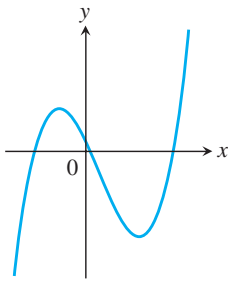
Thomas' Calculus, 14th SI edition • Section 4.4 • textbook pp. 228–232

Exercises 1, 4: Analyzing Functions from Graphs

Directions for Exercises 1–8.
Identify the inflection points and local maxima and minima of the functions graphed in Exercises 1–8. Identify the intervals on which the functions are concave up and concave down.

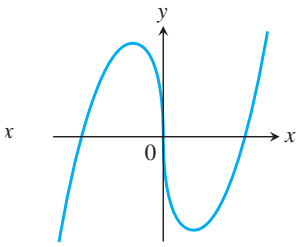
Exercise 1

$$y = \frac{x^3}{3} - \frac{x^2}{2} - 2x + \frac{1}{3}$$



Exercise 4

$$y = \frac{9}{14}x^{1/3}(x^2 - 7)$$



Exercises 17, 23, 32, 46: Graphing Functions

Directions for Exercises 9–50.
In Exercises 9–50, identify the coordinates of any local and absolute extreme points and inflection points. Graph the function.

Exercise 17

$$y = x^4 - 2x^2 = x^2(x^2 - 2)$$

Exercise 23

$$y = x + \sin x, \quad 0 \le x \le 2\pi$$

Exercise 32

$$y = \frac{\sqrt{1 - x^2}}{2x + 1}$$

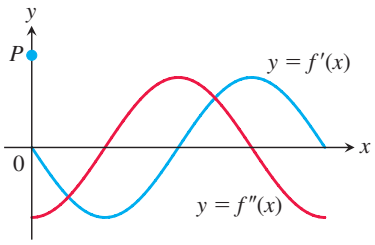
Exercise 46

$$y = |x^2 - 2x|$$

Exercise 75: Sketching from Graphs of and

Directions for Exercises 73–76.
Each of Exercises 73–76 shows the graphs of the first and second derivatives of a function $y = f(x)$. Copy the picture and add to it a sketch of the approximate graph of **f**, given that the graph passes through the point *P*.

Exercise 75



Exercise 78: Graphing Rational Functions

Directions for Exercises 77–94.
Graph the rational functions in Exercises 77–94 using all the steps in the graphing procedure on page 225.

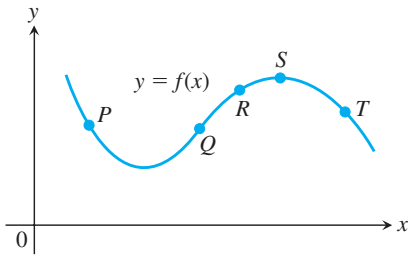
Exercise 78

$$y = \frac{x^2 - 49}{x^2 + 5x - 14}$$

Exercises 95, 97, 106, 107, 112: Theory and Examples

Exercise 95

The accompanying figure shows a portion of the graph of a twice-differentiable function $y = f(x)$. At each of the five labeled points, classify y' and y'' as positive, negative, or zero.



Exercise 97

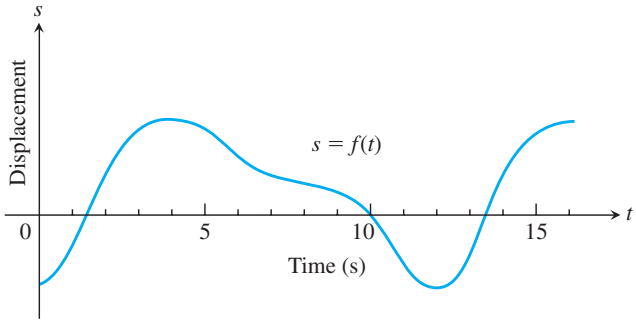
Sketch the graph of a twice-differentiable function $y = f(x)$ with the following properties. Label coordinates where possible.

x	y	Derivatives
$x < 2$		$y' < 0, \quad y'' > 0$
2	1	$y' = 0, \quad y'' > 0$
$2 < x < 4$		$y' > 0, \quad y'' > 0$
4	4	$y' > 0, \quad y'' = 0$
$4 < x < 6$		$y' > 0, \quad y'' < 0$
6	7	$y' = 0, \quad y'' < 0$
$x > 6$		$y' < 0, \quad y'' < 0$

Motion Along a Line The graphs in Exercises 105 and 106 show the position $s = f(t)$ of an object moving up and down on a coordinate line. **(a)** When is the object moving away from the origin? Toward the origin? At approximately what times is the **(b)** velocity equal to zero? **(c)** Acceleration equal to zero? **(d)** When is the acceleration positive? Negative?

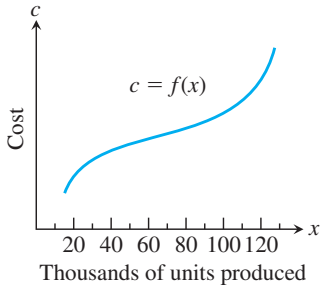
Directions for Exercises 105–106.

Exercise 106



Exercise 107

Marginal cost The accompanying graph shows the hypothetical cost $c = f(x)$ of manufacturing x items. At approximately what production level does the marginal cost change from decreasing to increasing?



Exercise 112

Can anything be said about the graph of a function $y = f(x)$ that has a continuous second derivative that is never zero? Give reasons for your answer.

4.5 Applied Optimization

Assigned: 1, 7, 12, 20, 22, 28, 42, 52, 53, 55, 58, 71 • Specified subparts are marked below.

Thomas' Calculus, 14th SI edition • Section 4.5 • textbook pp. 238–244

Exercises 1, 7, 12, 20, 22, 28: Mathematical Applications

Exercise 1

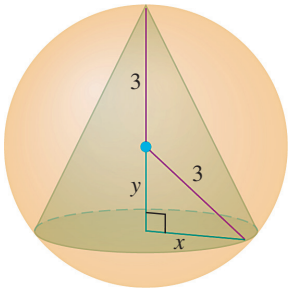
Minimizing perimeter What is the smallest perimeter possible for a rectangle whose area is 16 cm^2, and what are its dimensions?

Exercise 7

The best fencing plan A rectangular plot of farmland will be bounded on one side by a river and on the other three sides by a single-strand electric fence. With 800 m of wire at your disposal, what is the largest area you can enclose, and what are its dimensions?

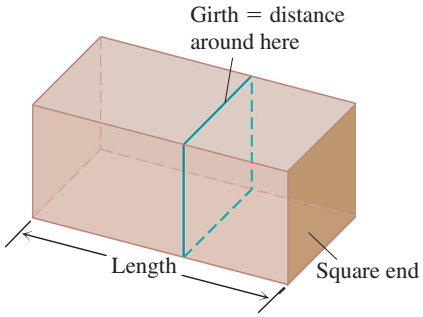
Exercise 12

Find the volume of the largest right circular cone that can be inscribed in a sphere of radius 3.



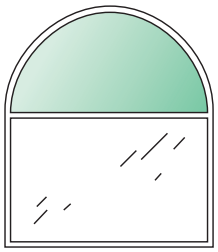
Exercise 20 • (a) only

a. A certain postal service will accept a box for domestic shipment only if the sum of its length and girth (distance around) does not exceed 276 cm. What dimensions will give a box with a square end the largest possible volume?



Exercise 22

A window is in the form of a rectangle surmounted by a semicircle. The rectangle is of clear glass, whereas the semicircle is of tinted glass that transmits only half as much light per unit area as clear glass does. The total perimeter is fixed. Find the proportions of the window that will admit the most light. Neglect the thickness of the frame.



Exercise 28

Find the point on the line $\frac{x}{a} + \frac{y}{b} = 1$ that is closest to the origin.

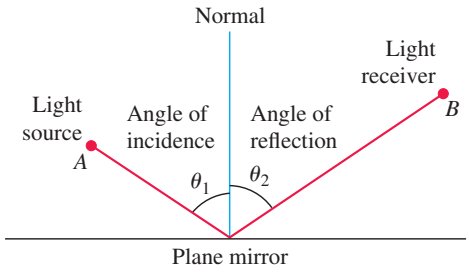
Exercises 42, 52, 53: Physical Applications

Exercise 42

Quickest route Jane is 2 km offshore in a boat and wishes to reach a coastal village 6 km down a straight shoreline from the point nearest the boat. She can row 2 km/h and can walk 5 km/h. Where should she land her boat to reach the village in the least amount of time?

Exercise 52

Fermat's principle in optics Light from a source A is reflected by a plane mirror to a receiver at point B, as shown in the accompanying figure. Show that for the light to obey Fermat's principle, the angle of incidence must equal the angle of reflection, both measured from the line normal to the reflecting surface. (This result can also be derived without calculus. There is a purely geometric argument, which you may prefer.)



Exercise 53

Tin pest When metallic tin is kept below 13.2°C, it slowly becomes brittle and crumbles to a gray powder. Tin objects eventually crumble to this gray powder spontaneously if kept in a cold climate for years. The Europeans who saw tin organ pipes in their churches crumble away years ago called the change *tin pest* because it seemed to be contagious, and indeed it was, for the gray powder is a catalyst for its own formation.

A *catalyst* for a chemical reaction is a substance that controls the rate of reaction without undergoing any permanent change in itself. An *autocatalytic reaction* is one whose product is a catalyst for its own formation. Such a reaction may proceed slowly at first if the amount of catalyst present is small and slowly again at the end, when most of the original substance is used up. But in between, when both the substance and its catalyst product are abundant, the reaction proceeds at a faster pace.

In some cases, it is reasonable to assume that the rate $v = dx/dt$ of the reaction is proportional both to the amount of the original substance present and to the amount of product. That is, v may be considered to be a function of x alone, and

$$v = kx(a - x) = kax - kx^2,$$

where

$x =$ the amount of product

$a =$ the amount of substance at the beginning

$k =$ a positive constant.

At what value of x does the rate v have a maximum? What is the maximum value of v?

Exercises 55, 58: Business and Economics

Exercise 55

It costs you c dollars each to manufacture and distribute backpacks. If the backpacks sell at x dollars each, the number sold is given by

$$n = \frac{a}{x - c} + b(100 - x),$$

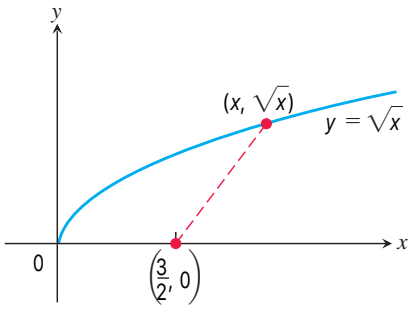
where a and b are positive constants. What selling price will bring a maximum profit?

Exercise 58

Production level Prove that the production level (if any) at which average cost is smallest is a level at which the average cost equals marginal cost.

Exercise 71 • (a) only

a. How close does the curve $y = \sqrt{x}$ come to the point $(3/2, 0)$? (*Hint:* If you minimize the *square* of the distance, you can avoid square roots.)



4.6 Newton's Method

Assigned: 1, 3, 9, 11, 12, 22

Thomas' Calculus, 14th SI edition • Section 4.6 • textbook pp. 247–249

Exercises 1, 3: Root Finding

Exercise 1

Use Newton's method to estimate the solutions of the equation $x^2 + x - 1 = 0$. Start with $x_0 = -1$ for the left-hand solution and with $x_0 = 1$ for the solution on the right. Then, in each case, find x_2.

Exercise 3

Use Newton's method to estimate the two zeros of the function $f(x) = x^4 + x - 3$. Start with $x_0 = -1$ for the left-hand zero and with $x_0 = 1$ for the zero on the right. Then, in each case, find x_2.

Exercise 9

Guessing a root Suppose that your first guess is lucky, in the sense that x_0 is a root of $f(x) = 0$. Assuming that $f'(x_0)$ is defined and not 0, what happens to x_1 and later approximations?

Exercises 11, 12, 22: Theory and Examples

Exercise 11

Oscillation Show that if $h > 0$, applying Newton's method to

$$f(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ \sqrt{-x}, & x < 0 \end{cases}$$

leads to $x_1 = -h$ if $x_0 = h$ and to $x_1 = h$ if $x_0 = -h$. Draw a picture that shows what is going on.

Exercise 12

Approximations that get worse and worse Apply Newton's method to $f(x) = x^{1/3}$ with $x_0 = 1$ and calculate x_1, x_2, x_3, and x_4. Find a formula for $|x_n|$. What happens to $|x_n|$ as $n \to \infty$? Draw a picture that shows what is going on.

Exercise 22

Intersection of curves At what value(s) of x does $\cos x = -x$?

4.7 Antiderivatives

Assigned: 1, 7, 12, 23, 32, 34, 52, 59, 63, 75, 85, 99, 103, 107

Thomas' Calculus, 14th SI edition • Section 4.7 • textbook pp. 255–259

Exercises 1, 7, 12: Finding Antiderivatives

Directions for Exercises 1–16.
In Exercises 1–16, find an antiderivative for each function. Do as many as you can mentally. Check your answers by differentiation.

Exercise 1

a. $2x$ **b.** x^2 **c.** $x^2 - 2x + 1$

Exercise 7

a. $\frac{3}{2}\sqrt{x}$ **b.** $\frac{1}{2\sqrt{x}}$ **c.** $\sqrt{x} + \frac{1}{\sqrt{x}}$

Exercise 12

a. $\pi \cos \pi x$ **b.** $\frac{\pi}{2}\cos\frac{\pi x}{2}$ **c.** $\cos\frac{\pi x}{2} + \pi \cos x$

Exercises 23, 32, 34, 52: Finding Indefinite Integrals

Directions for Exercises 17–56.
In Exercises 17–56, find the most general antiderivative or indefinite integral. You may need to try a solution and then adjust your guess. Check your answers by differentiation.

Exercise 23

$$\int \left(\frac{1}{x^2} - x^2 - \frac{1}{3}\right) dx$$

Exercise 32

$$\int x^{-3}(x + 1)\, dx$$

Exercise 34

$$\int \frac{4 + \sqrt{t}}{t^3} dt$$

Exercise 52

$$\int (2 + \tan^2 \theta)\, d\theta$$

Exercises 59, 63: Checking Antiderivative Formulas

Directions for Exercises 57–62.
Verify the formulas in Exercises 57–62 by differentiation.

Exercise 59

$$\int \sec^2 (5x - 1)\, dx = \frac{1}{5}\tan (5x - 1) + C$$

Exercise 63

Right, or wrong? Say which for each formula and give a brief reason for each answer.

a. $\int x \sin x\, dx = \frac{x^2}{2}\sin x + C$

b. $\int x \sin x\, dx = -x \cos x + C$

c. $\int x \sin x\, dx = -x \cos x + \sin x + C$

Exercises 75, 85: Initial Value Problems

Directions for Exercises 71–90.
Give reasons for your answer.

Solve the initial value problems in Exercises 71–90.

Exercise 75

$$\frac{dy}{dx} = 3x^{-2/3}, \quad y(-1) = -5$$

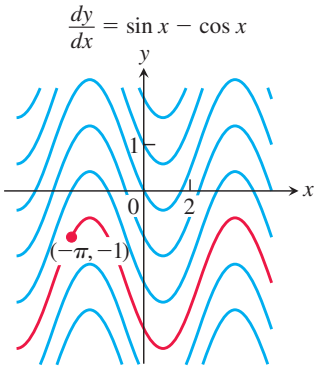
Exercise 85

$$\frac{d^2r}{dt^2} = \frac{2}{t^3}; \quad \left.\frac{dr}{dt}\right|_{t=1} = 1, \quad r(1) = 1$$

Exercise 99: Solution (Integral) Curves

Directions for Exercises 97–100.
Exercises 97–100 show solution curves of differential equations. In each exercise, find an equation for the curve through the labeled point.

Exercise 99



Exercises 103, 107: Applications

Exercise 103

Stopping a car in time You are driving along a highway at a steady 108 km/h (30 m/s) when you see an accident ahead and slam on the brakes. What constant deceleration is required to stop your car in 75 m? To find out, carry out the following steps.

1. Solve the initial value problem

   Differential equation: $\frac{d^2s}{dt^2} = -k$ (k constant)

   Initial conditions: $\frac{ds}{dt} = 30$ and $s = 0$ when $t = 0$.

   Measuring time and distance from when the brakes are applied

2. Find the value of t that makes $ds/dt = 0$. (The answer will involve k.)
3. Find the value of k that makes $s = 75$ for the value of t you found in Step 2.

Exercise 107

Motion with constant acceleration The standard equation for the position s of a body moving with a constant acceleration a along a coordinate line is

$$s = \frac{a}{2}t^2 + v_0 t + s_0, \tag{1}$$

where v_0 and s_0 are the body's velocity and position at time $t = 0$. Derive this equation by solving the initial value problem

Differential equation: $\frac{d^2s}{dt^2} = a$

Initial conditions: $\frac{ds}{dt} = v_0$ and $s = s_0$ when $t = 0$.

5.1 Area and Estimating with Finite Sums

Assigned: 1, 3, 5, 7, 19

Thomas' Calculus, 14th SI edition • Section 5.1 • textbook pp. 274–276

Exercises 1, 3, 5, 7: Area

Directions for Exercises 1–4.
n Exercises 1–4, use finite approximations to estimate the area under he graph of the function using

a. a lower sum with two rectangles of equal width.

b. a lower sum with four rectangles of equal width.

c. an upper sum with two rectangles of equal width.

d. an upper sum with four rectangles of equal width.

Exercise 1

$f(x) = x^2$ between $x = 0$ and $x = 1$.

Exercise 3

$f(x) = 1/x$ between $x = 1$ and $x = 5$.

Directions for Exercises 5–8.
Using rectangles each of whose height is given by the value of he function at the midpoint of the rectangle's base (*the midpoint rule*), stimate the area under the graphs of the following functions, using rst two and then four rectangles.

Exercise 5

$f(x) = x^2$ between $x = 0$ and $x = 1$.

Exercise 7

$f(x) = 1/x$ between $x = 1$ and $x = 5$.

Exercise 19

Water pollution Oil is leaking out of a tanker damaged at sea. The damage to the tanker is worsening as evidenced by the increased leakage each hour, recorded in the following table.

Time (h)	0	1	2	3	4
Leakage (L / h)	50	70	97	136	190

Time (h)	5	6	7	8
Leakage (L / h)	265	369	516	720

a. Give an upper and a lower estimate of the total quantity of oil that has escaped after 5 hours.

b. Repeat part (a) for the quantity of oil that has escaped after 8 hours.

c. The tanker continues to leak 720 L / h after the first 8 hours. If the tanker originally contained 25,000 L of oil, approximately how many more hours will elapse in the worst case before all the oil has spilled? In the best case?

5.2 Sigma Notation and Limits of Finite Sums

Assigned: 8, 12, 30, 32, 37, 39, 41–44

Thomas' Calculus, 14th SI edition • Section 5.2 • textbook pp. 282–283

Exercises 8, 12: Sigma Notation

Exercise 8

Which of the following express $1 - 2 + 4 - 8 + 16 - 32$ in sigma notation?

a. $\sum_{k=1}^{6} (-2)^{k-1}$ **b.** $\sum_{k=0}^{5} (-1)^k 2^k$ **c.** $\sum_{k=-2}^{3} (-1)^{k+1} 2^{k+2}$

Directions for Exercises 11–16.
Express the sums in Exercises 11–16 in sigma notation. The form of your answer will depend on your choice for the starting index.

Exercise 12

$1 + 4 + 9 + 16$

Exercises 30, 32: Values of Finite Sums

Directions for Exercises 19–32.
Evaluate the sums in Exercises 19–32.

Exercise 30

a. $\sum_{k=9}^{36} k$ **b.** $\sum_{k=3}^{17} k^2$ **c.** $\sum_{k=18}^{71} k(k-1)$

Exercise 32

a. $\sum_{k=1}^{n} \left(\frac{1}{n} + 2n\right)$ **b.** $\sum_{k=1}^{n} \frac{c}{n}$ **c.** $\sum_{k=1}^{n} \frac{k}{n^2}$

Exercises 37, 39, 41, 42: Riemann Sums

Directions for Exercises 37–42.
In Exercises 37–42, graph each function $f(x)$ over the given interval. Partition the interval into four subintervals of equal length. Then add to your sketch the rectangles associated with the Riemann sum $\Sigma_{k=1}^{4} f(c_k)\,\Delta x_k$, given that c_k is the **(a)** left-hand endpoint, **(b)** right-hand endpoint, **(c)** midpoint of the kth subinterval. (Make a separate sketch for each set of rectangles.)

Exercise 37

$f(x) = x^2 - 1, \quad [0, 2]$

Exercise 39

$f(x) = \sin x, \quad [-\pi, \pi]$

Exercise 41

Find the norm of the partition $P = \{0, 1.2, 1.5, 2.3, 2.6, 3\}$.

Exercise 42

Find the norm of the partition $P = \{-2, -1.6, -0.5, 0, 0.8, 1\}$.

Exercises 43, 44: Limits of Riemann Sums

Directions for Exercises 43–50.
For the functions in Exercises 43–50, find a formula for the Riemann sum obtained by dividing the interval $[a, b]$ into n equal subintervals and using the right-hand endpoint for each c_k. Then take a limit of these sums as $n \to \infty$ to calculate the area under the curve over $[a, b]$.

Exercise 43

$f(x) = 1 - x^2$ over the interval $[0, 1]$.

Exercise 44

$f(x) = 2x$ over the interval $[0, 3]$.

5.3 The Definite Integral

Assigned: 1, 3, 6, 8, 9, 17, 20, 55, 71, 73, 76–78, 81, 83

Thomas' Calculus, 14th SI edition • Section 5.3 • textbook pp. 292–296

Exercises 1, 3, 6, 8: Interpreting Limits of Sums as Integrals

Directions for Exercises 1–8.
Express the limits in Exercises 1–8 as definite integrals.

Exercise 1

$$\lim_{\|P\|\to 0} \sum_{k=1}^{n} c_k^{\,2}\, \Delta x_k, \text{ where } P \text{ is a partition of } [\,0, 2\,]$$

Exercise 3

$$\lim_{\|P\|\to 0} \sum_{k=1}^{n} (c_k^{\,2} - 3c_k)\, \Delta x_k, \text{ where } P \text{ is a partition of } [\,-7, 5\,]$$

Exercise 6

$$\lim_{\|P\|\to 0} \sum_{k=1}^{n} \sqrt{4 - c_k^{\,2}}\, \Delta x_k, \text{ where } P \text{ is a partition of } [\,0, 1\,]$$

Exercise 8

$$\lim_{\|P\|\to 0} \sum_{k=1}^{n} (\tan c_k)\, \Delta x_k, \text{ where } P \text{ is a partition of } [\,0, \pi/4\,]$$

Exercise 9

Suppose that **f** and g are integrable and that

$$\int_1^2 f(x)\,dx = -4, \quad \int_1^5 f(x)\,dx = 6, \quad \int_1^5 g(x)\,dx = 8.$$

Use the rules in Table 5.6 to find

a. $\displaystyle\int_2^2 g(x)\,dx$

b. $\displaystyle\int_5^1 g(x)\,dx$

c. $\displaystyle\int_1^2 3f(x)\,dx$

d. $\displaystyle\int_2^5 f(x)\,dx$

e. $\displaystyle\int_1^5 [\,f(x) - g(x)\,]\,dx$

f. $\displaystyle\int_1^5 [\,4f(x) - g(x)\,]\,dx$

Exercises 17, 20: Using Known Areas to Find Integrals

Directions for Exercises 15–22.
In Exercises 15–22, graph the integrands and use known area formulas to evaluate the integrals.

Exercise 17

$$\int_{-3}^{3} \sqrt{9 - x^2}\,dx$$

Exercise 20

$$\int_{-1}^{1} (1 - |x|)\,dx$$

Exercise 55: Finding Average Value

Directions for Exercises 55–62.
In Exercises 55–62, graph the function and find its average value over the given interval.

Exercise 55

$$f(x) = x^2 - 1 \quad \text{on} \quad [\,0, \sqrt{3}\,]$$

Exercises 71, 73, 76–78, 81, 83: Theory and Examples

Exercise 71

What values of a and b maximize the value of

$$\int_a^b (x - x^2)\,dx?$$

(*Hint:* Where is the integrand positive?)

Exercise 73

Use the Max-Min Inequality to find upper and lower bounds for the value of

$$\int_0^1 \frac{1}{1 + x^2}\,dx.$$

Exercise 76

Show that the value of $\int_0^1 \sqrt{x + 8}\,dx$ lies between $2\sqrt{2} \approx 2.8$ and 3.

Exercise 77

Integrals of nonnegative functions Use the Max-Min Inequality to show that if $\mathbf{f}$ is integrable then

$$f(x) \geq 0 \quad \text{on} \quad [a, b] \quad \Rightarrow \quad \int_a^b f(x)\, dx \geq 0.$$

Exercise 78

Integrals of nonpositive functions Show that if $\mathbf{f}$ is integrable then

$$f(x) \leq 0 \quad \text{on} \quad [a, b] \quad \Rightarrow \quad \int_a^b f(x)\, dx \leq 0.$$

Exercise 81

If $\text{av}(f)$ really is a typical value of the integrable function $f(x)$ on $[a, b]$, then the constant function $\text{av}(f)$ should have the same integral over $[a, b]$ as $\mathbf{f}$. Does it? That is, does

$$\int_a^b \text{av}(f)\, dx = \int_a^b f(x)\, dx?$$

Give reasons for your answer.

Exercise 83

Upper and lower sums for increasing functions

a. Suppose the graph of a continuous function $f(x)$ rises steadily as x moves from left to right across an interval $[a, b]$. Let P be a partition of $[a, b]$ into n subintervals of equal length

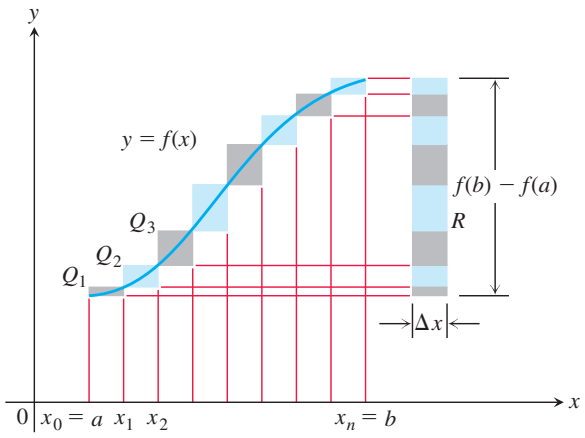
Exercise 83 (continued)

$\Delta x = (b - a)/n$. Show by referring to the accompanying figure that the difference between the upper and lower sums for $\mathbf{f}$ on this partition can be represented graphically as the area of a rectangle R whose dimensions are $[f(b) - f(a)]$ by Δx. (*Hint:* The difference $U - L$ is the sum of areas of rectangles whose diagonals $Q_0Q_1, Q_1Q_2, \ldots, Q_{n-1}Q_n$ lie approximately along the curve. There is no overlapping when these rectangles are shifted horizontally onto R.)

b. Suppose that instead of being equal, the lengths Δx_k of the subintervals of the partition of $[a, b]$ vary in size. Show that

$$U - L \leq |f(b) - f(a)|\, \Delta x_{\max},$$

where $\Delta x_{\max}$ is the norm of P, and hence that $\lim_{\|P\| \to 0} (U - L) = 0$.



5.4 The Fundamental Theorem of Calculus

Assigned: 1, 3, 4, 8–10, 12, 17, 22, 24, 27, 28, 30, 40, 44, 46, 61, 63, 68, 71, 73

Thomas' Calculus, 14th SI edition • Section 5.4 • textbook pp. 305–307

Exercises 1, 3, 4, 8–10, 12, 17, 22, 24, 27, 28, 30: Evaluating Integrals

Directions for Exercises 1–28.
Evaluate the integrals in Exercises 1–28.

Exercise 1

$$\int_0^2 x(x - 3)\,dx$$

Exercise 3

$$\int_{-2}^2 \frac{3}{(x + 3)^4}\,dx$$

Exercise 4

$$\int_{-1}^1 x^{299}\,dx$$

Exercise 8

$$\int_1^{32} x^{-6/5}\,dx$$

Exercise 9

$$\int_0^{\pi/3} 2\sec^2 x\,dx$$

Exercise 10

$$\int_0^{\pi} (1 + \cos x)\,dx$$

Exercise 12

$$\int_0^{\pi/3} 4\,\frac{\sin u}{\cos^2 u}\,du$$

Exercise 17

$$\int_0^{\pi/8} \sin 2x\,dx$$

Exercise 22

$$\int_{-3}^{-1} \frac{y^5 - 2y}{y^3}\,dy$$

Exercise 24

$$\int_1^8 \frac{(x^{1/3} + 1)(2 - x^{2/3})}{x^{1/3}}\,dx$$

Exercise 27

$$\int_{-4}^4 |x|\,dx$$

Exercise 28

$$\int_0^{\pi} \frac{1}{2}(\cos x + |\cos x|)\,dx$$

Directions for Exercises 29–32.
In Exercises 29–32, guess an antiderivative for the integrand function. Validate your guess by differentiation and then evaluate the given definite integral. (*Hint:* Keep the Chain Rule in mind when trying to guess an antiderivative. You will learn how to find such antiderivatives in the next section.)

Exercise 30

$$\int_1^{\pi^2} \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$$

Exercises 40, 44, 46: Derivatives of Integrals

Directions for Exercises 39–46.
Find dy/dx in Exercises 39–46.

Exercise 40

$$y = \int_1^x \frac{1}{t}dt, \quad x > 0$$

Exercise 44

$$y = \left(\int_0^x (t^3 + 1)^{10}\, dt\right)^3$$

Exercise 46

$$y = \int_{\tan x}^{0} \frac{dt}{1 + t^2}$$

Exercises 61, 63, 68, 71, 73: Theory and Examples

Exercise 61

Archimedes' area formula for parabolic arches Archimedes (287–212 b.c.), inventor, military engineer, physicist, and the greatest mathematician of classical times in the Western world, discovered that the area under a parabolic arch is two-thirds the base times the height. Sketch the parabolic arch $y = h - (4h/b^2)x^2$, $-b/2 \le x \le b/2$, assuming that h and b are positive. Then use calculus to find the area of the region enclosed between the arch and the x-axis.

Exercise 63

Cost from marginal cost The marginal cost of printing a poster when x posters have been printed is

$$\frac{dc}{dx} = \frac{1}{2\sqrt{x}}$$

dollars. Find $c(100) - c(1)$, the cost of printing posters 2–100.

Exercise 68

Find $f(4)$ if $\int_0^x f(t)\, dt = x \cos \pi x$.

Exercise 71

Suppose that f has a positive derivative for all values of x and that $f(1) = 0$. Which of the following statements must be true of the function

$$g(x) = \int_0^x f(t)\, dt?$$

Give reasons for your answers.

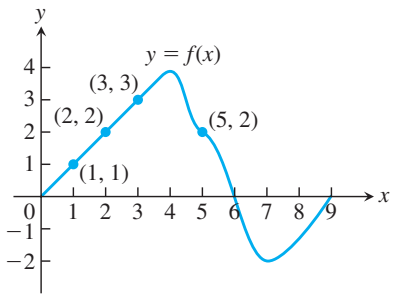
a. g is a differentiable function of x.
b. g is a continuous function of x.
c. The graph of g has a horizontal tangent at $x = 1$.
d. g has a local maximum at $x = 1$.
e. g has a local minimum at $x = 1$.
f. The graph of g has an inflection point at $x = 1$.
g. The graph of dg/dx crosses the x-axis at $x = 1$.

Exercise 73

Suppose that f is the differentiable function shown in the accompanying graph and that the position at time t (s) of a particle moving along a coordinate axis is

$$s = \int_0^t f(x)\, dx$$

meters. Use the graph to answer the following questions. Give reasons for your answers.



a. What is the particle's velocity at time $t = 5$?
b. Is the acceleration of the particle at time $t = 5$ positive, or negative?
c. What is the particle's position at time $t = 3$?
d. At what time during the first 9 s does s have its largest value?
e. Approximately when is the acceleration zero?
f. When is the particle moving toward the origin? Away from the origin?
g. On which side of the origin does the particle lie at time $t = 9$?

5.5 Indefinite Integrals and the Substitution Method

Assigned: 18, 20–22, 26, 31, 38, 42, 43, 46, 56, 62

Thomas' Calculus, 14th SI edition • Section 5.5 • textbook pp. 313–314

Exercises 18, 20–22, 26, 31, 38, 42, 43, 46: Evaluating Indefinite Integrals

Directions for Exercises 17–50.
Evaluate the integrals in Exercises 17–50.

Exercise 18

$$\int \frac{1}{\sqrt{5s + 4}}\, ds$$

Exercise 20

$$\int 3y\sqrt{7 - 3y^2}\, dy$$

Exercise 21

$$\int \frac{1}{\sqrt{x}\left(1 + \sqrt{x}\right)^2}\, dx$$

Exercise 22

$$\int \sqrt{\sin x}\cos^3 x\, dx$$

Exercise 26

$$\int \tan^7 \frac{x}{2}\sec^2 \frac{x}{2} dx$$

Exercise 31

$$\int \frac{\sin(2t + 1)}{\cos^2(2t + 1)} dt$$

Exercise 38

$$\int \sqrt{\frac{x - 1}{x^5}} dx$$

Exercise 42

$$\int \sqrt{\frac{x^4}{x^3 - 1}} dx$$

Exercise 43

$$\int x(x - 1)^{10}\, dx$$

Exercise 46

$$\int (x + 5)(x - 5)^{1/3}\, dx$$

Exercises 56, 62: Initial Value Problems

Directions for Exercises 55–60.
Solve the initial value problems in Exercises 55–60.

Exercise 56

$$\frac{dy}{dx} = 4x\,(x^2 + 8)^{-1/3}, \quad y(0) = 0$$

Exercise 62

The acceleration of a particle moving back and forth on a line is $a = d^2s/dt^2 = \pi^2 \cos \pi t$ m/s^2 for all t. If $s = 0$ and $v = 8$ m/s when $t = 0$, find s when $t = 1$ s.

5.6 Definite Integral Substitutions and the Area Between Curves

Assigned: 8, 11, 12, 21, 23, 25, 59, 63, 71, 72, 78, 82–87

Thomas' Calculus, 14th SI edition • Section 5.6 • textbook pp. 321–324

Exercises 8, 11, 12, 21, 23: Evaluating Definite Integrals

Directions for Exercises 1–24.
Use the Substitution Formula in Theorem 7 to evaluate the integrals in Exercises 1–24.

Exercise 8

a. $\displaystyle\int_0^1 \frac{10\sqrt{v}}{(1+v^{3/2})^2}\,dv$ **b.** $\displaystyle\int_1^4 \frac{10\sqrt{v}}{(1+v^{3/2})^2}\,dv$

Exercise 11

a. $\displaystyle\int_0^1 t\sqrt{4+5t}\,dt$ **b.** $\displaystyle\int_1^9 t\sqrt{4+5t}\,dt$

Exercise 12

a. $\displaystyle\int_0^{\pi/6} (1-\cos 3t)\sin 3t\,dt$

b. $\displaystyle\int_{\pi/6}^{\pi/3} (1-\cos 3t)\sin 3t\,dt$

Exercise 21

$$\int_0^1 (4y - y^2 + 4y^3 + 1)^{-2/3}(12y^2 - 2y + 4)\,dy$$

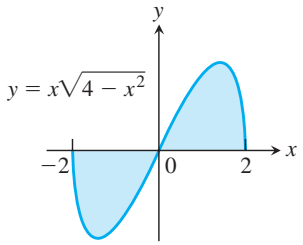
Exercise 23

$$\int_0^{\sqrt[3]{\pi^2}} \sqrt{\theta}\cos^2(\theta^{3/2})\,d\theta$$

Exercises 25, 59, 63: Area

Directions for Exercises 25–40.
Find the total areas of the shaded regions in Exercises 25–40.

Exercise 25



Directions for Exercises 59–62.
Find the areas of the regions enclosed by the curves in Exercises 59–62.

Exercise 59

$4x^2 + y = 4$ and $x^4 - y = 1$

Directions for Exercises 63–70.
Find the areas of the regions enclosed by the lines and curves in Exercises 63–70.

Exercise 63

$y = 2\sin x$ and $y = \sin 2x$, $0 \le x \le \pi$

Exercises 71, 72, 78, 82: Area Between Curves

Exercise 71

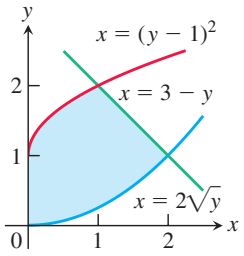
Find the area of the propeller-shaped region enclosed by the curve $x - y^3 = 0$ and the line $x - y = 0$.

Exercise 72

Find the area of the propeller-shaped region enclosed by the curves $x - y^{1/3} = 0$ and $x - y^{1/5} = 0$.

Exercise 78

Find the area of the region in the first quadrant bounded on the left by the y-axis, below by the curve $x = 2\sqrt{y}$, above left by the curve $x = (y - 1)^2$, and above right by the line $x = 3 - y$.



Exercise 82

True, sometimes true, or never true? The area of the region between the graphs of the continuous functions $y = f(x)$ and $y = g(x)$ and the vertical lines $x = a$ and $x = b$ $(a < b)$ is

$$\int_a^b [\, f(x) - g(x) \,]\, dx.$$

Give reasons for your answer.

Exercises 83–87: Theory and Examples

Exercise 83

Suppose that $F(x)$ is an antiderivative of $f(x) = (\sin x)/x$, $x > 0$. Express

$$\int_1^3 \frac{\sin 2x}{x} dx$$

in terms of F.

Exercise 84

Show that if f is continuous, then

$$\int_0^1 f(x)\, dx = \int_0^1 f(1 - x)\, dx.$$

Exercise 85

Suppose that

$$\int_0^1 f(x)\, dx = 3.$$

Find

$$\int_{-1}^0 f(x)\, dx$$

if **a.** f is odd, **b.** f is even.

Exercise 86

a. Show that if f is odd on $[-a, a]$, then

$$\int_{-a}^a f(x)\, dx = 0.$$

b. Test the result in part (a) with $f(x) = \sin x$ and $a = \pi/2$.

Exercise 87

If f is a continuous function, find the value of the integral

$$I = \int_0^a \frac{f(x)\, dx}{f(x) + f(a - x)}$$

by making the substitution $u = a - x$ and adding the resulting integral to I.

6.1 Volumes Using Cross-Sections

Assigned: 1, 10, 12, 14, 21, 25, 29, 31, 39, 43, 45, 53, 61

Thomas' Calculus, 14th SI edition • Section 6.1 • textbook pp. 339–343

Exercises 1, 10, 12, 14: Volumes by Slicing

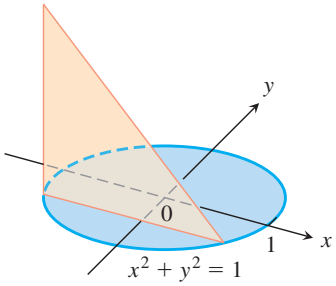
Directions for Exercises 1–10.
Find the volumes of the solids in Exercises 1–10.

Exercise 1

The solid lies between planes perpendicular to the x-axis at $x = 0$ and $x = 4$. The cross-sections perpendicular to the axis on the interval $0 \le x \le 4$ are squares whose diagonals run from the parabola $y = -\sqrt{x}$ to the parabola $y = \sqrt{x}$.

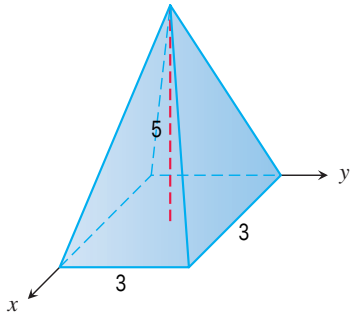
Exercise 10

The base of the solid is the disk $x^2 + y^2 \le 1$. The cross-sections by planes perpendicular to the y-axis between $y = -1$ and $y = 1$ are isosceles right triangles with one leg in the disk.



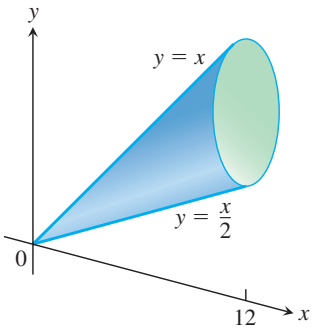
Exercise 12

Find the volume of the given pyramid, which has a square base of area 9 and height 5.



Exercise 14

Cavalieri's principle A solid lies between planes perpendicular to the x-axis at $x = 0$ and $x = 12$. The cross-sections by planes perpendicular to the x-axis are circular disks whose diameters run from the line $y = x/2$ to the line $y = x$ as shown in the accompanying figure. Explain why the solid has the same volume as a right circular cone with base radius 3 and height 12.



Exercises 21, 25, 29, 31: Volumes by the Disk Method

Directions for Exercises 21–26.
Find the volumes of the solids generated by revolving the regions bounded by the lines and curves in Exercises 21–26 about the x-axis.

Exercise 21

$y = x^2, \quad y = 0, \quad x = 2$

Exercise 25

$y = \sqrt{\cos x}, \quad 0 \le x \le \pi/2, \quad y = 0, \quad x = 0$

Directions for Exercises 29–34.
Find the volumes of the solids generated by revolving the regions bounded by the lines and curves in Exercises 29–34 about the y-axis.

Exercise 29

The region enclosed by $x = \sqrt{5}y^2$, $x = 0$, $y = -1$, $y = 1$

Exercise 31

The region enclosed by $x = \sqrt{2\sin 2y}$, $0 \le y \le \pi/2$, $x = 0$

Exercises 39, 43, 45: Volumes by the Washer Method

Directions for Exercises 37–42.
Find the volumes of the solids generated by revolving the regions bounded by the lines and curves in Exercises 37–42 about the x-axis.

Exercise 39

$y = x^2 + 1$, $y = x + 3$

Directions for Exercises 43–46.
In Exercises 43–46, find the volume of the solid generated by revolving each region about the y-axis.

Exercise 43

The region enclosed by the triangle with vertices (1, 0), (2, 1), and (1, 1)

Exercise 45

The region in the first quadrant bounded above by the parabola $y = x^2$, below by the x-axis, and on the right by the line $x = 2$

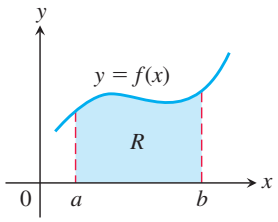
Exercises 53, 61: Theory and Applications

Exercise 53

The volume of a torus The disk $x^2 + y^2 \le a^2$ is revolved about the line $x = b$ ($b > a$) to generate a solid shaped like a doughnut and called a *torus*. Find its volume. (*Hint:* $\int_{-a}^{a} \sqrt{a^2 - y^2}\, dy = \pi a^2/2$, since it is the area of a semicircle of radius a.)

Exercise 61

Consider the region R bounded by the graphs of $y = f(x) > 0$, $x = a > 0$, $x = b > a$, and $y = 0$ (see accompanying figure). If the volume of the solid formed by revolving R about the x-axis is 4π, and the volume of the solid formed by revolving R about the line $y = -1$ is 8π, find the area of R.



6.3 Arc Length

Assigned: 1, 2, 9, 23, 30

Thomas' Calculus, 14th SI edition • Section 6.3 • textbook pp. 355–356

Exercises 1, 2, 9: Finding Lengths of Curves

Directions for Exercises 1–12.
Find the lengths of the curves in Exercises 1–12. If you have graphing software, you may want to graph these curves to see what they look like.

Exercise 1

$y = (1/3)(x^2 + 2)^{3/2}$ from $x = 0$ to $x = 3$

Exercise 2

$y = x^{3/2}$ from $x = 0$ to $x = 4$

Exercise 9

$$y = \frac{x^3}{3} + \frac{1}{4x}, \quad 1 \le x \le 3$$

Exercises 23, 30: Theory and Examples

Exercise 23

Find the length of the curve

$$y = \int_0^x \sqrt{\cos 2t}\, dt$$

from $x = 0$ to $x = \pi/4$.

Exercise 30

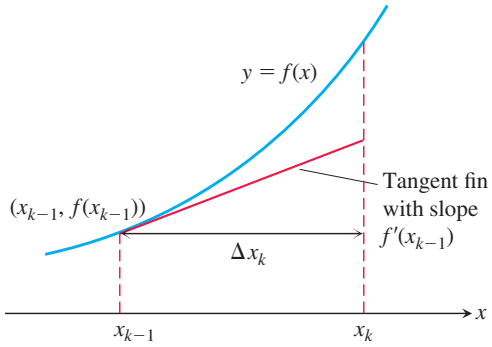
Using tangent fins to derive the length formula for curves Assume that f is smooth on $[a, b]$ and partition the interval $[a, b]$ in the usual way. In each subinterval $[x_{k-1}, x_k]$, construct the *tangent fin* at the point $(x_{k-1}, f(x_{k-1}))$, as shown in the accompanying figure.

a. Show that the length of the kth tangent fin over the interval $[x_{k-1}, x_k]$ equals $\sqrt{(\Delta x_k)^2 + (f'(x_{k-1})\,\Delta x_k)^2}$.

b. Show that

$$\lim_{n\to\infty} \sum_{k=1}^{n} (\text{length of } k\text{th tangent fin}) = \int_a^b \sqrt{1 + (f'(x))^2}\, dx,$$

which is the length L of the curve $y = f(x)$ from a to b.



6.4 Areas of Surfaces of Revolution

Assigned: 13, 18, 22, 23, 25, 28

Thomas' Calculus, 14th SI edition • Section 6.4 • textbook pp. 360–362

Exercises 13, 18, 22, 23, 25, 28: Finding Surface Area

Directions for Exercises 13–23.
Exercises 13–23 about the indicated axes. If you have a grapher, you may want to graph these curves to see what they look like.

Exercise 13

$y = x^3/9, \quad 0 \le x \le 2; \quad x$-axis

Exercise 18

$x = (1/3)y^{3/2} - y^{1/2}, \quad 1 \le y \le 3; \quad y$-axis

Exercise 22

$y = (1/3)(x^2 + 2)^{3/2}, \quad 0 \le x \le \sqrt{2}; \quad y$-axis (*Hint:* Express $ds = \sqrt{dx^2 + dy^2}$ in terms of dx, and evaluate the integral $S = \int 2\pi x\, ds$ with appropriate limits.)

Exercise 23

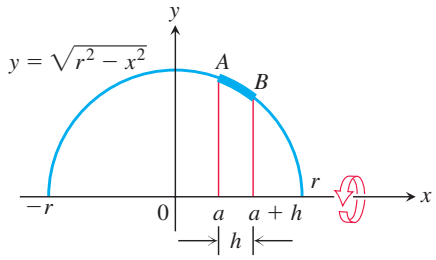
$x = (y^4/4) + 1/(8y^2), \quad 1 \le y \le 2; \quad x$-axis (*Hint:* Express $ds = \sqrt{dx^2 + dy^2}$ in terms of dy, and evaluate the integral $S = \int 2\pi y\, ds$ with appropriate limits.)

Exercise 25

Testing the new definition Show that the surface area of a sphere of radius a is still $4\pi a^2$ by using Equation (3) to find the area of the surface generated by revolving the curve $y = \sqrt{a^2 - x^2}$, $-a \le x \le a$, about the x-axis.

Exercise 28

Slicing bread Did you know that if you cut a spherical loaf of bread into slices of equal width, each slice will have the same amount of crust? To see why, suppose the semicircle $y = \sqrt{r^2 - x^2}$ shown here is revolved about the x-axis to generate a sphere. Let AB be an arc of the semicircle that lies above an interval of length h on the x-axis. Show that the area swept out by AB does not depend on the location of the interval. (It does depend on the length of the interval.)



6.5 Work and Fluid Forces

Assigned: 3, 9, 12, 16, 25, 26, 31, 38, 46

Thomas' Calculus, 14th SI edition • Section 6.5 • textbook pp. 367–371

Exercise 3

Spring constant It took 1800 J of work to stretch a spring from its natural length of 2 m to a length of 5 m. Find the spring's force constant.

Exercises 9, 12: Work Done by a Variable Force

Exercise 9

Lifting a rope A mountain climber is about to haul up a 50-m length of hanging rope. How much work will it take if the rope weighs 0.624 N/m?

Exercise 12

Force of attraction When a particle of mass m is at $(x, 0)$, it is attracted toward the origin with a force whose magnitude is k/x^2. If the particle starts from rest at $x = b$ and is acted on by no other forces, find the work done on it by the time it reaches $x = a$, $0 < a < b$.

Exercise 16

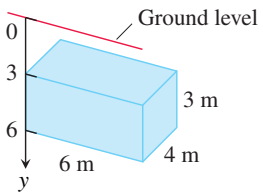
Emptying a cistern The rectangular cistern (storage tank for rainwater) shown has its top 3 m below ground level. The cistern, currently full, is to be emptied for inspection by pumping its contents to ground level.

a. How much work will it take to empty the cistern?

b. How long will it take a 1/2-hp pump, rated at 370 W, to pump the tank dry?

c. How long will it take the pump in part (b) to empty the tank halfway? (It will be less than half the time required to empty the tank completely.)

d. **The weight of water** What are the answers to parts (a) through (c) in a location where water weighs 9780 N/m^3? 9820 N/m^3?



Exercises 25, 26, 31: Work and Kinetic Energy

Exercise 25

Kinetic energy If a variable force of magnitude $F(x)$ moves an object of mass m along the x-axis from x_1 to x_2, the object's velocity v can be written as dx/dt (where t represents time). Use Newton's second law of motion $F = m(dv/dt)$ and the Chain Rule

$$\frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$$

to show that the net work done by the force in moving the object from x_1 to x_2 is

$$W = \int_{x_1}^{x_2} F(x)\,dx = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2,$$

where v_1 and v_2 are the object's velocities at x_1 and x_2. In physics, the expression $(1/2)mv^2$ is called the *kinetic energy* of an object of mass m moving with velocity v. Therefore, *the work done by the force equals the change in the object's kinetic energy*, and we can find the work by calculating this change.

Directions for Exercises 26–30.
In Exercises 26–30, use the result of Exercise 25.

Exercise 26

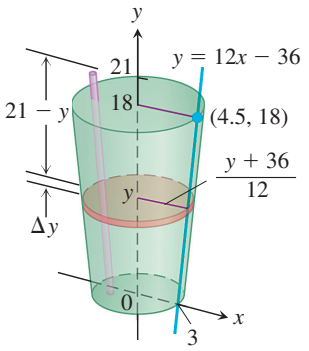
Tennis A 60 g tennis ball was served at 50 m/s (180 km/h). How much work was done on the ball to make it go this fast?

Exercise 31

Drinking a milkshake The truncated conical container shown here is full of strawberry milkshake which has a density of 0.8 g/cm^3. As you can see, the container is 18 cm deep, 6 cm across at the base, and 9 cm across at the top (a standard size at Brigham's in Boston). The straw sticks up 3 cm above the top.

Exercise 31 (continued)

About how much work does it take to suck up the milkshake through the straw (neglecting friction)?

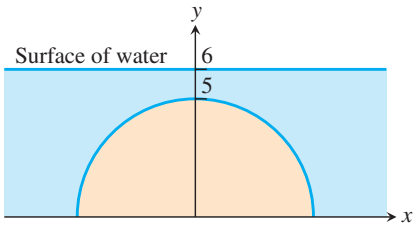


Dimensions in centimeters

Exercises 38, 46: Finding Fluid Forces

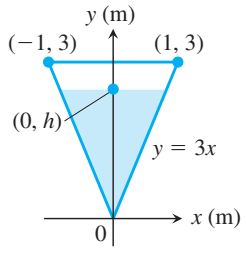
Exercise 38

Semicircular plate Calculate the fluid force on one side of a semicircular plate of radius 5 m that rests vertically on its diameter at the bottom of a pool filled with water to a depth of 6 m.

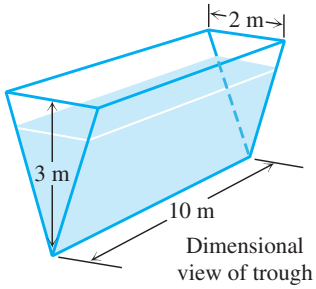


Exercise 46

The end plates of the trough shown here were designed to withstand a fluid force of 25,000 N. How many cubic meters of water can the tank hold without exceeding this limitation? Round down to the nearest cubic meter. What is the value of h?



End view of trough



Dimensional view of trough

7.1 Inverse Functions and Their Derivatives

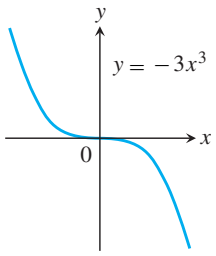
Assigned: 1–6, 12, 14, 17, 25, 31, 38, 39, 44, 56, 59

Thomas' Calculus, 14th SI edition • Section 7.1 • textbook pp. 394–396

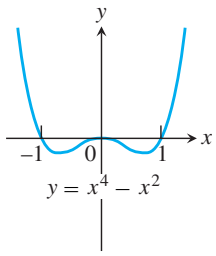
Exercises 1–6: Identifying One-to-One Functions Graphically

Directions for Exercises 1–6.
Which of the functions graphed in Exercises 1–6 are one-to-one, and which are not?

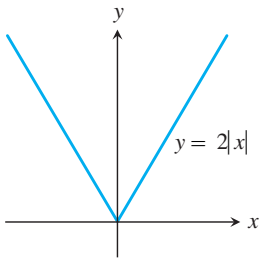
Exercise 1



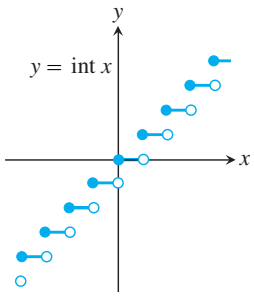
Exercise 2



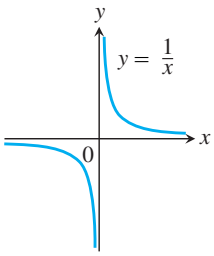
Exercise 3



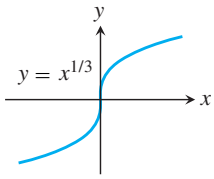
Exercise 4



Exercise 5



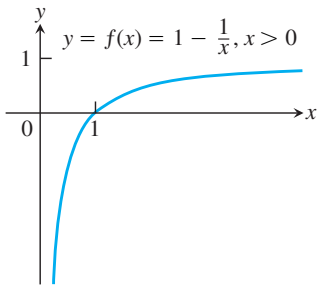
Exercise 6



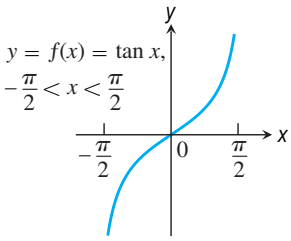
Exercises 12, 14, 17: Graphing Inverse Functions

Directions for Exercises 11–16.
Each of Exercises 11–16 shows the graph of a function $y = f(x)$. Copy the graph and draw in the line $y = x$. Then use symmetry with respect to the line $y = x$ to add the graph of f^{-1} to your sketch. (It is not necessary to find a formula for f^{-1}.) Identify the domain and range of f^{-1}.

Exercise 12



Exercise 14



Exercise 17

a. Graph the function $f(x) = \sqrt{1 - x^2}$, $0 \le x \le 1$. What symmetry does the graph have?

b. Show that f is its own inverse. (Remember that $\sqrt{x^2} = x$ if $x \ge 0$.)

Exercises 25, 31, 38, 39, 44: Derivatives of Inverse Functions

Directions for Exercises 25–34.
Each of Exercises 25–34 gives a formula for a function $y = f(x)$. In each case, find $f^{-1}(x)$ and identify the domain and range of f^{-1}. As a check, show that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$.

Exercise 25

$f(x) = x^5$

Exercise 31

$f(x) = \dfrac{x + 3}{x - 2}$

Directions for Exercises 35–38.
In Exercises 35–38:

a. Find $f^{-1}(x)$.

b. Graph f and f^{-1} together.

c. Evaluate df/dx at $x = a$ and df^{-1}/dx at $x = f(a)$ to show that at these points $df^{-1}/dx = 1/(df/dx)$.

Exercise 38

$f(x) = 2x^2, \quad x \ge 0, \quad a = 5$

Exercise 39

a. Show that $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverses of one another.

b. Graph f and g over an x-interval large enough to show the graphs intersecting at (1, 1) and (−1, −1). Be sure the picture shows the required symmetry about the line $y = x$.

c. Find the slopes of the tangent lines to the graphs of f and g at (1, 1) and (−1, −1) (four tangent lines in all).

d. What lines are tangent to the curves at the origin?

Exercise 44

Suppose that the differentiable function $y = g(x)$ has an inverse and that the graph of g passes through the origin with slope 2. Find the slope of the graph of g^{-1} at the origin.

Exercises 56, 59: Theory and Applications

Exercise 56

If $f(x)$ is one-to-one and $f(x)$ is never zero, can anything be said about $h(x) = 1/f(x)$? Is it also one-to-one? Give reasons for your answer.

Exercise 59

Assume that f and g are differentiable functions that are inverses of one another so that $(g \circ f)(x) = x$. Differentiate both sides of this equation with respect to x using the Chain Rule to express $(g \circ f)'(x)$ as a product of derivatives of g and f. What do you find? (This is not a proof of Theorem 1 because we assume here the theorem's conclusion that $g = f^{-1}$ is differentiable.)

7.2 Natural Logarithms

Assigned: 1, 4, 7, 10, 11, 23, 25, 36, 41, 44, 47, 55, 58, 67, 72, 77, 81, 89

Thomas' Calculus, 14th SI edition • Section 7.2 • textbook pp. 403–404

Exercises 1, 4: Using the Algebraic Properties—Theorem 2

Exercise 1

Express the following logarithms in terms of $\ln 2$ and $\ln 3$.

a. $\ln 0.75$ **b.** $\ln (4/9)$ **c.** $\ln (1/2)$

d. $\ln \sqrt[3]{9}$ **e.** $\ln 3\sqrt{2}$ **f.** $\ln \sqrt{13.5}$

Directions for Exercises 3–4.
Use the properties of logarithms to simplify the expressions in Exercises 3 and 4.

Exercise 4

a. $\ln \sec\theta + \ln \cos\theta$ **b.** $\ln (8x + 4) - 2 \ln 2$

c. $3 \ln \sqrt[3]{t^2 - 1} - \ln (t + 1)$

In Exercises 5 and 6, solve for t.

Exercises 7, 10, 11, 23, 25, 36: Finding Derivatives

Directions for Exercises 7–38.
In Exercises 7–38, find the derivative of y with respect to x, t, or θ, as appropriate.

Exercise 7

$$y = \ln 3x$$

Exercise 10

$$y = \ln (t^{3/2})$$

Exercise 11

$$y = \ln \frac{3}{x}$$

Exercise 23

$$y = \frac{\ln x}{1 + \ln x}$$

Exercise 25

$$y = \ln (\ln x)$$

Exercise 36

$$y = \ln \sqrt{\frac{(x + 1)^5}{(x + 2)^{20}}}$$

Exercises 41, 44, 47, 55: Evaluating Integrals

Directions for Exercises 39–56.
Evaluate the integrals in Exercises 39–56.

Exercise 41

$$\int \frac{2y\, dy}{y^2 - 25}$$

Exercise 44

$$\int_0^{\pi/3} \frac{4 \sin\theta}{1 - 4 \cos\theta}\, d\theta$$

Exercise 47

$$\int_2^4 \frac{dx}{x (\ln x)^2}$$

Exercise 55

$$\int \frac{dx}{2\sqrt{x} + 2x}$$

Exercises 58, 67: Logarithmic Differentiation

Directions for Exercises 57–70.
In Exercises 57–70, use logarithmic differentiation to find the derivative of y with respect to the given independent variable.

Exercise 58

$$y = \sqrt{(x^2 + 1)(x - 1)^2}$$

Exercise 67

$$y = \frac{x\sqrt{x^2 + 1}}{(x + 1)^{2/3}}$$

Exercise 72

a. Prove that $f(x) = x - \ln x$ is increasing for $x > 1$.

b. Using part (a), show that $\ln x < x$ if $x > 1$.

Exercises 77, 81, 89: Identifying Extrema

Exercise 77

The region in the first quadrant bounded by the coordinate axes, the line $y = 3$, and the curve $x = 2/\sqrt{y + 1}$ is revolved about the y-axis to generate a solid. Find the volume of the solid.

Exercise 81

Find the lengths of the following curves.

a. $y = (x^2/8) - \ln x, \quad 4 \le x \le 8$

b. $x = (y/4)^2 - 2 \ln (y/4), \quad 4 \le y \le 12$

Exercise 89

The linearization of $\ln (1 + x)$ at $x = 0$ Instead of approximating $\ln x$ near $x = 1$, we approximate $\ln (1 + x)$ near $x = 0$. We get a simpler formula this way.

a. Derive the linearization $\ln (1 + x) \approx x$ at $x = 0$.

b. Estimate to five decimal places the error involved in replacing $\ln (1 + x)$ by x on the interval $[0, 0.1]$.

c. Graph $\ln (1 + x)$ and x together for $0 \le x \le 0.5$. Use different colors, if available. At what points does the approximation of $\ln (1 + x)$ seem best? Least good? By reading coordinates from the graphs, find as good an upper bound for the error as your grapher will allow.

7.3 Exponential Functions

Assigned: 1, 8, 16, 18, 26, 28, 29, 33, 42, 52, 60, 63, 66, 69, 78, 79, 86, 89, 92, 95, 100, 103, 115, 122, 131, 133, 141, 146

Thomas' Calculus, 14th SI edition • Section 7.3 • textbook pp. 412–415

Exercise 1: Solving Exponential Equations

Directions for Exercises 1–4.
In Exercises 1–4, solve for t.

Exercise 1

a. $e^{-0.3t} = 27$ **b.** $e^{kt} = \frac{1}{2}$ **c.** $e^{(\ln 0.2)t} = 0.4$

Exercises 8, 16, 18, 26, 28, 29: Finding Derivatives

Directions for Exercises 7–26.
In Exercises 7–26, find the derivative of y with respect to x, t, or θ, as appropriate.

Exercise 8

$$y = e^{2x/3}$$

Exercise 16

$$y = \ln(3\theta e^{-\theta})$$

Exercise 18

$$y = \theta^3 e^{-2\theta} \cos 5\theta$$

Exercise 26

$$y = \int_{e^{4\sqrt{x}}}^{e^{2x}} \ln t \, dt$$

Directions for Exercises 27–32.
In Exercises 27–32, find dy/dx.

Exercise 28

$$\ln xy = e^{x+y}$$

Exercise 29

$$e^{2x} = \sin(x + 3y)$$

Exercises 33, 42, 52: Finding Integrals

Directions for Exercises 33–54.
Evaluate the integrals in Exercises 33–54.

Exercise 33

$$\int (e^{3x} + 5e^{-x})\, dx$$

Exercise 42

$$\int \frac{e^{-\sqrt{r}}}{\sqrt{r}} dr$$

Exercise 52

$$\int_0^{\sqrt{\ln \pi}} 2x\, e^{x^2} \cos(e^{x^2})\, dx$$

Exercises 60, 63, 66, 69, 78, 79, 86: Differentiation

Directions for Exercises 59–86.
In Exercises 59–86, find the derivative of y with respect to the given independent variable.

Exercise 60

$$y = 3^{-x}$$

Exercise 63

$$y = x^{\pi}$$

Exercise 66

$$y = (\ln \theta)^{\pi}$$

Exercise 69

$$y = 2^{\sin 3t}$$

Exercise 78

$$y = \log_5 \sqrt{\left(\frac{7x}{3x+2}\right)^{\ln 5}}$$

Exercise 79

$$y = \theta \sin(\log_7 \theta)$$

Exercise 86

$$y = t \log_3\left(e^{(\sin t)(\ln 3)}\right)$$

Exercises 89, 92, 95, 100, 103: Integration

Directions for Exercises 87–96.
Evaluate the integrals in Exercises 87–96.

Exercise 89

$$\int_0^1 2^{-\theta}\, d\theta$$

Exercise 92

$$\int_1^4 \frac{2\sqrt{x}}{\sqrt{x}}\, dx$$

Exercise 95

$$\int_2^4 x^{2x}(1 + \ln x)\, dx$$

Directions for Exercises 97–110.
Evaluate the integrals in Exercises 97–110.

Exercise 100

$$\int_1^e x^{(\ln 2)-1}\, dx$$

Exercise 103

$$\int_1^4 \frac{\ln 2 \log_2 x}{x}\, dx$$

Exercises 115, 122: Logarithmic Differentiation

Directions for Exercises 115–122.
In Exercises 115–122, use logarithmic differentiation to find the derivative of y with respect to the given independent variable.

Exercise 115

$$y = (x + 1)^x$$

Exercise 122

$$y = (\ln x)^{\ln x}$$

Exercises 131, 133, 141, 146: Theory and Applications

Exercise 131

Let $f(x) = xe^{-x}$.

a. Find all absolute extreme values for f.

b. Find all inflection points for f.

Exercise 133

Find the absolute maximum value of $f(x) = x^2 \ln(1/x)$ and say where it is assumed.

Directions for Exercises 139–142.
In Exercises 139–142, find the length of each curve.

Exercise 141

$y = \ln(\cos x)$ from $x = 0$ to $x = \pi/4$

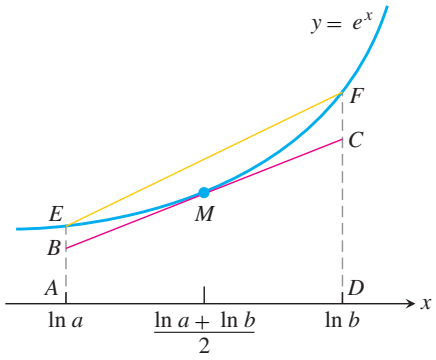
Exercise 146

The geometric, logarithmic, and arithmetic mean inequality

a. Show that the graph of e^x is concave up over every interval of x-values.

b. Show, by reference to the accompanying figure, that if $0 < a < b$, then

$$e^{(\ln a + \ln b)/2} \cdot (\ln b - \ln a) < \int_{\ln a}^{\ln b} e^x\, dx < \frac{e^{\ln a} + e^{\ln b}}{2} \cdot (\ln b - \ln a).$$



NOT TO SCALE

c. Use the inequality in part (b) to conclude that

$$\sqrt{ab} < \frac{b - a}{\ln b - \ln a} < \frac{a + b}{2}.$$

This inequality says that the geometric mean of two positive numbers is less than their logarithmic mean, which in turn is less than their arithmetic mean.

7.5 Indeterminate Forms and L'Hôpital's Rule

Assigned: 16, 20, 21, 26, 31, 38, 40, 41, 45, 51, 53, 59–61, 66–68, 75, 85, 88

Thomas' Calculus, 14th SI edition • Section 7.5 • textbook pp. 432–434

Exercises 16, 20, 21, 26, 31, 38, 40, 41, 45: Applying l'Hôpital's Rule

Directions for Exercises 7–50.
Use l'Hôpital's rule to find the limits in Exercises 7–50.

Exercise 16

$$\lim_{x\to 0} \frac{\sin x - x}{x^3}$$

Exercise 20

$$\lim_{x\to 1} \frac{x - 1}{\ln x - \sin \pi x}$$

Exercise 21

$$\lim_{x\to 0} \frac{x^2}{\ln (\sec x)}$$

Exercise 26

$$\lim_{x\to (\pi/2)^-} \left(\frac{\pi}{2} - x\right) \tan x$$

Exercise 31

$$\lim_{x\to \infty} \frac{\ln (x + 1)}{\log_2 x}$$

Exercise 38

$$\lim_{x\to 0^+} (\ln x - \ln \sin x)$$

Exercise 40

$$\lim_{x\to 0^+} \left(\frac{3x + 1}{x} - \frac{1}{\sin x}\right)$$

Exercise 41

$$\lim_{x\to 1^+} \left(\frac{1}{x - 1} - \frac{1}{\ln x}\right)$$

Exercise 45

$$\lim_{t\to \infty} \frac{e^t + t^2}{e^t - t}$$

Exercises 51, 53, 59–61, 66: Indeterminate Powers and Products

Directions for Exercises 51–66.
Find the limits in Exercises 51–66.

Exercise 51

$$\lim_{x\to 1^+} x^{1/(1-x)}$$

Exercise 53

$$\lim_{x\to \infty} (\ln x)^{1/x}$$

Exercise 59

$$\lim_{x\to 0^+} x^x$$

Exercise 60

$$\lim_{x\to 0^+} \left(1 + \frac{1}{x}\right)^x$$

Exercise 61

$$\lim_{x\to \infty} \left(\frac{x + 2}{x - 1}\right)^x$$

Exercise 66

$$\lim_{x\to 0^+} \sin x \cdot \ln x$$

Exercises 67, 68, 75, 85, 88: Theory and Applications

Directions for Exercises 67–74.
L'Hôpital's Rule does not help with the limits in Exercises 67–74. Try it—you just keep on cycling. Find the limits some other way.

Exercise 67

$$\lim_{x\to \infty} \frac{\sqrt{9x + 1}}{\sqrt{x + 1}}$$

Exercise 68

$$\lim_{x\to 0^+} \frac{\sqrt{x}}{\sqrt{\sin x}}$$

Exercise 75

Which one is correct, and which one is wrong? Give reasons for your answers.

a. $\lim_{x\to 3} \dfrac{x-3}{x^2-3} = \lim_{x\to 3} \dfrac{1}{2x} = \dfrac{1}{6}$ **b.** $\lim_{x\to 3} \dfrac{x-3}{x^2-3} = \dfrac{0}{6} = 0$

Exercise 85

Show that

$$\lim_{k\to\infty} \left(1 + \frac{r}{k}\right)^k = e^r.$$

Exercise 88

Find $f'(0)$ for $f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$

7.8 Relative Rates of Growth

Assigned: 1, 9–12, 20–22 • Specified subparts are marked below.
Thomas' Calculus, 14th SI edition • Section 7.8 • textbook pp. 458–459

Exercise 1

Which of the following functions grow faster than e^x as $x \to \infty$? Which grow at the same rate as e^x? Which grow slower?

a. $x - 3$ **b.** $x^3 + \sin^2 x$
c. $\sqrt{x}$ **d.** 4^x
e. $(3/2)^x$ **f.** $e^{x/2}$
g. $e^x/2$ **h.** $\log_{10} x$

Exercises 9–12: Big-oh and Little-oh; Order

Exercise 9

True, or false? As $x \to \infty$,

a. $x = o(x)$ **b.** $x = o(x + 5)$
c. $x = O(x + 5)$ **d.** $x = O(2x)$
e. $e^x = o(e^{2x})$ **f.** $x + \ln x = O(x)$
g. $\ln x = o(\ln 2x)$ **h.** $\sqrt{x^2 + 5} = O(x)$

Exercise 10

True, or false? As $x \to \infty$,

a. $\dfrac{1}{x+3} = O\left(\dfrac{1}{x}\right)$ **b.** $\dfrac{1}{x} + \dfrac{1}{x^2} = O\left(\dfrac{1}{x}\right)$
c. $\dfrac{1}{x} - \dfrac{1}{x^2} = o\left(\dfrac{1}{x}\right)$ **d.** $2 + \cos x = O(2)$
e. $e^x + x = O(e^x)$ **f.** $x \ln x = o(x^2)$
g. $\ln(\ln x) = O(\ln x)$ **h.** $\ln(x) = o(\ln(x^2 + 1))$

Exercise 11

Show that if positive functions $f(x)$ and $g(x)$ grow at the same rate as $x \to \infty$, then $f = O(g)$ and $g = O(f)$.

Exercise 12

When is a polynomial $f(x)$ of smaller order than a polynomial $g(x)$ as $x \to \infty$? Give reasons for your answer.

Exercises 20–22: Other Comparisons

Exercise 20

The function e^x outgrows any polynomial Show that e^x grows faster as $x \to \infty$ than any polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

Exercise 21 • (a) only

a. Show that $\ln x$ grows slower as $x \to \infty$ than $x^{1/n}$ for any positive integer n, even $x^{1/1{,}000{,}000}$.

Exercise 22

The function $\ln x$ grows slower than any polynomial Show that $\ln x$ grows slower as $x \to \infty$ than any nonconstant polynomial.

7.6 Inverse Trigonometric Functions

Assigned: 1, 3, 5, 7, 13, 16, 18, 22, 25, 33, 48, 51, 56, 67, 69, 73, 78, 91, 97, 98, 102, 103, 129, 133

Thomas' Calculus, 14th SI edition • Section 7.6 • textbook pp. 442–446

Exercises 1, 3, 5, 7: Common Values

Directions for Exercises 1–8.
Use reference triangles like those in Examples 1 and 3 to find the angles in Exercises 1–8.

Exercise 1

a. $\tan^{-1} 1$ **b.** $\arctan\left(-\sqrt{3}\right)$ **c.** $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Exercise 3

a. $\sin^{-1}\left(\frac{-1}{2}\right)$ **b.** $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ **c.** $\arcsin\left(\frac{-\sqrt{3}}{2}\right)$

Exercise 5

a. $\cos^{-1}\left(\frac{1}{2}\right)$ **b.** $\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$ **c.** $\arccos\left(\frac{\sqrt{3}}{2}\right)$

Exercise 7

a. $\operatorname{arcsec}\left(-\sqrt{2}\right)$ **b.** $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ **c.** $\sec^{-1}(-2)$

Exercises 13, 16, 18: Limits

Directions for Exercises 13–20.
Find the limits in Exercises 13–20. (If in doubt, look at the function's graph.)

Exercise 13

$$\lim_{x \to 1^-} \sin^{-1} x$$

Exercise 16

$$\lim_{x \to -\infty} \tan^{-1} x$$

Exercise 18

$$\lim_{x \to -\infty} \sec^{-1} x$$

Exercises 22, 25, 33: Finding Derivatives

Directions for Exercises 21–42.
In Exercises 21–42, find the derivative of y with respect to the appropriate variable.

Exercise 22

$$y = \cos^{-1}(1/x)$$

Exercise 25

$$y = \sec^{-1}(2s + 1)$$

Exercise 33

$$y = \ln(\tan^{-1} x)$$

Exercises 48, 51, 56, 67, 69, 73, 78, 91: Evaluating Integrals

Directions for Exercises 47–70.
Evaluate the integrals in Exercises 47–70.

Exercise 48

$$\int \frac{dx}{\sqrt{1 - 4x^2}}$$

Exercise 51

$$\int \frac{dx}{x\sqrt{25x^2 - 2}}$$

Exercise 56

$$\int_{-2}^{2} \frac{dt}{4 + 3t^2}$$

Exercise 67

$$\int_{0}^{\ln \sqrt{3}} \frac{e^x\, dx}{1 + e^{2x}}$$

Exercise 69

$$\int \frac{y\, dy}{\sqrt{1 - y^4}}$$

Directions for Exercises 71–84.
Evaluate the integrals in Exercises 71–84.

Exercise 73

$$\int_{-1}^{0} \frac{6\,dt}{\sqrt{3 - 2t - t^2}}$$

Exercise 78

$$\int_{2}^{4} \frac{2\,dx}{x^2 - 6x + 10}$$

Directions for Exercises 85–94.
Evaluate the integrals in Exercises 85–94.

Exercise 91

$$\int_{\sqrt{2}}^{2} \frac{\sec^2(\sec^{-1} x)\,dx}{x\sqrt{x^2 - 1}}$$

Exercises 97, 98, 102, 103: L'Hôpital's Rule

Directions for Exercises 97–104.
Find the limits in Exercises 97–104.

Exercise 97

$$\lim_{x \to 0} \frac{\sin^{-1} 5x}{x}$$

Exercise 98

$$\lim_{x \to 1^+} \frac{\sqrt{x^2 - 1}}{\sec^{-1} x}$$

Exercise 102

$$\lim_{x \to \infty} \frac{e^x \tan^{-1} e^x}{e^{2x} + x}$$

Exercise 103

$$\lim_{x \to 0^+} \frac{\left(\tan^{-1} \sqrt{x}\right)^2}{x\sqrt{x + 1}}$$

Exercises 129, 133: Applications and Theory

Directions for Exercises 129–130.
Find the volumes of the solids in Exercises 129 and 130.

Exercise 129

The solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis are

a. circles whose diameters stretch from the curve $y = -1/\sqrt{1 + x^2}$ to the curve $y = 1/\sqrt{1 + x^2}$.

b. vertical squares whose base edges run from the curve $y = -1/\sqrt{1 + x^2}$ to the curve $y = 1/\sqrt{1 + x^2}$.

Directions for Exercises 133–135.
In Exercises 133–135, find the domain and range of each composite function. Then graph the compositions on separate screens. Do the graphs make sense in each case? Give reasons for your answers. Comment on any differences you see.

Exercise 133

a. $y = \tan^{-1}(\tan x)$ **b.** $y = \tan(\tan^{-1} x)$

8.2 Integration by Parts

Assigned: 2, 6, 12, 17, 23, 28, 29, 31, 32, 37, 43, 45, 50, 51, 53, 65, 67, 75

Thomas' Calculus, 14th SI edition • Section 8.2 • textbook pp. 475–478

Exercises 2, 6, 12, 17, 23: Integration by Parts

Directions for Exercises 1–24.
Evaluate the integrals in Exercises 1–24 using integration by parts.

Exercise 2

$$\int \theta \cos \pi\theta \, d\theta$$

Exercise 6

$$\int_1^e x^3 \ln x \, dx$$

Exercise 12

$$\int \sin^{-1} y \, dy$$

Exercise 17

$$\int (x^2 - 5x)e^x \, dx$$

Exercise 23

$$\int e^{2x} \cos 3x \, dx$$

Exercises 28, 29: Using Substitution

Directions for Exercises 25–30.
Evaluate the integrals in Exercises 25–30 by using a substitution prior to integration by parts.

Exercise 28

$$\int \ln (x + x^2) \, dx$$

Exercise 29

$$\int \sin (\ln x) \, dx$$

Exercises 31, 32, 37, 43, 45, 50, 51, 53: Evaluating Integrals

Directions for Exercises 31–56.
Evaluate the integrals in Exercises 31–56. Some integrals do not require integration by parts.

Exercise 31

$$\int x \sec x^2 \, dx$$

Exercise 32

$$\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$$

Exercise 37

$$\int x^3 e^{x^4} \, dx$$

Exercise 43

$$\int \sqrt{x} \ln x \, dx$$

Exercise 45

$$\int \cos \sqrt{x} \, dx$$

Exercise 50

$$\int_0^{1/\sqrt{2}} 2x \sin^{-1} (x^2) \, dx$$

Exercise 51

$$\int x \tan^{-1} x \, dx$$

Exercise 53

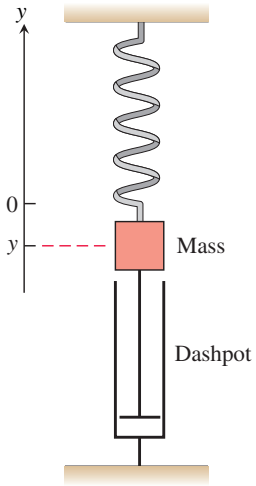
$$\int (1 + 2x^2)e^{x^2} \, dx$$

Exercise 65

Average value A retarding force, symbolized by the dashpot in the accompanying figure, slows the motion of the weighted spring so that the mass's position at time t is

$$y = 2e^{-t}\cos t, \qquad t \geq 0.$$

Find the average value of y over the interval $0 \leq t \leq 2\pi$.



Exercises 67, 75: Reduction Formulas

Directions for Exercises 67–71.
In Exercises 67–71, use integration by parts to establish the reduction formula.

Exercise 67

$$\int x^n \cos x\,dx = x^n \sin x - n\int x^{n-1}\sin x\,dx$$

Exercise 75

Show that

$$\int_a^b \left(\int_x^b f(t)\,dt\right) dx = \int_a^b (x-a)f(x)\,dx.$$

8.3 Trigonometric Integrals

Assigned: 3, 14, 15, 19, 20, 24, 27, 30, 33–35, 53, 54, 56, 64, 67

Thomas' Calculus, 14th SI edition • Section 8.3 • textbook pp. 483–484

Exercises 3, 14, 15, 19, 20: Powers of Sines and Cosines

Directions for Exercises 1–22.
Evaluate the integrals in Exercises 1–22.

Exercise 3

$$\int \cos^3 x \sin x \, dx$$

Exercise 14

$$\int_0^{\pi/2} \sin^2 x \, dx$$

Exercise 15

$$\int_0^{\pi/2} \sin^7 y \, dy$$

Exercise 19

$$\int 16 \sin^2 x \cos^2 x \, dx$$

Exercise 20

$$\int_0^{\pi} 8 \sin^4 y \cos^2 y \, dy$$

Exercises 24, 27, 30: Integrating Square Roots

Directions for Exercises 23–32.
Evaluate the integrals in Exercises 23–32.

Exercise 24

$$\int_0^{\pi} \sqrt{1 - \cos 2x} \, dx$$

Exercise 27

$$\int_{\pi/3}^{\pi/2} \frac{\sin^2 x}{\sqrt{1 - \cos x}} \, dx$$

Exercise 30

$$\int_{\pi/2}^{3\pi/4} \sqrt{1 - \sin 2x} \, dx$$

Exercises 33–35: Powers of Tangents and Secants

Directions for Exercises 33–50.
Evaluate the integrals in Exercises 33–50.

Exercise 33

$$\int \sec^2 x \tan x \, dx$$

Exercise 34

$$\int \sec x \tan^2 x \, dx$$

Exercise 35

$$\int \sec^3 x \tan x \, dx$$

Exercises 53, 54, 56: Products of Sines and Cosines

Directions for Exercises 51–56.
Evaluate the integrals in Exercises 51–56.

Exercise 53

$$\int_{-\pi}^{\pi} \sin 3x \sin 3x \, dx$$

Exercise 54

$$\int_0^{\pi/2} \sin x \cos x \, dx$$

Exercise 56

$$\int_{-\pi/2}^{\pi/2} \cos x \cos 7x \, dx$$

Exercises 64, 67: Assorted Integrations

Directions for Exercises 63–68.
Use any method to evaluate the integrals in Exercises 63–68.

Exercise 64

$$\int \frac{\sin^3 x}{\cos^4 x}\,dx$$

Exercise 67

$$\int x \sin^2 x\,dx$$

8.4 Trigonometric Substitutions

Assigned: 2, 3, 5, 8–10, 15, 17, 21, 26, 36, 37, 61, 62

Thomas' Calculus, 14th SI edition • Section 8.4 • textbook pp. 488–489

Exercises 2, 3, 5, 8–10: Using Trigonometric Substitutions

Directions for Exercises 1–14.
Evaluate the integrals in Exercises 1–14.

Exercise 2

$$\int \frac{3\,dx}{\sqrt{1 + 9x^2}}$$

Exercise 3

$$\int_{-2}^{2} \frac{dx}{4 + x^2}$$

Exercise 5

$$\int_{0}^{3/2} \frac{dx}{\sqrt{9 - x^2}}$$

Exercise 8

$$\int \sqrt{1 - 9t^2}\,dt$$

Exercise 9

$$\int \frac{dx}{\sqrt{4x^2 - 49}}, \quad x > \frac{7}{2}$$

Exercise 10

$$\int \frac{5\,dx}{\sqrt{25x^2 - 9}}, \quad x > \frac{3}{5}$$

Exercises 15, 17, 21, 26, 36, 37: Assorted Integrations

Directions for Exercises 15–34.
Use any method to evaluate the integrals in Exercises 15–34. Most will require trigonometric substitutions, but some can be evaluated by other methods.

Exercise 15

$$\int \frac{x}{\sqrt{9 - x^2}}\,dx$$

Exercise 17

$$\int \frac{x^3\,dx}{\sqrt{x^2 + 4}}$$

Exercise 21

$$\int \sqrt{\frac{x + 1}{1 - x}}\,dx$$

Exercise 26

$$\int \frac{x^2\,dx}{(x^2 - 1)^{5/2}}, \quad x > 1$$

Directions for Exercises 35–48.
In Exercises 35–48, use an appropriate substitution and then a trigonometric substitution to evaluate the integrals.

Exercise 36

$$\int_{\ln(3/4)}^{\ln(4/3)} \frac{e^t\,dt}{(1 + e^{2t})^{3/2}}$$

Exercise 37

$$\int_{1/12}^{1/4} \frac{2\,dt}{\sqrt{t} + 4t\sqrt{t}}$$

Exercises 61, 62: Applications and Examples

Exercise 61

Evaluate $\int x^3 \sqrt{1 - x^2}\, dx$ using

a. integration by parts.

b. a u-substitution.

c. a trigonometric substitution.

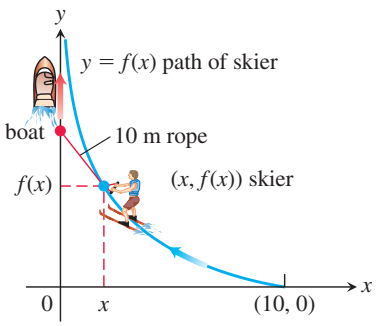
Exercise 62

Path of a water skier Suppose that a boat is positioned at the origin with a water skier tethered to the boat at the point (10, 0) on a rope 10 m long. As the boat travels along the positive y-axis, the skier is pulled behind the boat along an unknown path $y = f(x)$, as shown in the accompanying figure.

a. Show that $f'(x) = \dfrac{-\sqrt{100 - x^2}}{x}$.

(*Hint*: Assume that the skier is always pointed directly at the boat and the rope is on a line tangent to the path $y = f(x)$.)

b. Solve the equation in part (a) for $f(x)$, using $f(10) = 0$.



NOT TO SCALE

8.5 Integration of Rational Functions by Partial Fractions

Assigned: 6, 7, 13, 16, 20, 23, 27, 31, 34, 35, 41

Thomas' Calculus, 14th SI edition • Section 8.5 • textbook pp. 495–497

Exercises 6, 7: Expanding Quotients into Partial Fractions

Directions for Exercises 1–8.
Expand the quotients in Exercises 1–8 by partial fractions.

Exercise 6

$$\frac{z}{z^3 - z^2 - 6z}$$

Exercise 7

$$\frac{t^2 + 8}{t^2 - 5t + 6}$$

Exercises 13, 16: Nonrepeated Linear Factors

Directions for Exercises 9–16.
In Exercises 9–16, express the integrand as a sum of partial fractions and evaluate the integrals.

Exercise 13

$$\int_4^8 \frac{y\,dy}{y^2 - 2y - 3}$$

Exercise 16

$$\int \frac{x + 3}{2x^3 - 8x}\,dx$$

Exercise 20: Repeated Linear Factors

Directions for Exercises 17–20.
In Exercises 17–20, express the integrand as a sum of partial fractions and evaluate the integrals.

Exercise 20

$$\int \frac{x^2\,dx}{(x - 1)(x^2 + 2x + 1)}$$

Exercises 23, 27, 31: Irreducible Quadratic Factors

Directions for Exercises 21–32.
In Exercises 21–32, express the integrand as a sum of partial fractions and evaluate the integrals.

Exercise 23

$$\int \frac{y^2 + 2y + 1}{(y^2 + 1)^2}\,dy$$

Exercise 27

$$\int \frac{x^2 - x + 2}{x^3 - 1}\,dx$$

Exercise 31

$$\int \frac{2\theta^3 + 5\theta^2 + 8\theta + 4}{(\theta^2 + 2\theta + 2)^2}\,d\theta$$

Exercises 34, 35: Improper Fractions

Directions for Exercises 33–38.
In Exercises 33–38, perform long division on the integrand, write the proper fraction as a sum of partial fractions, and then evaluate the integral.

Exercise 34

$$\int \frac{x^4}{x^2 - 1}\,dx$$

Exercise 35

$$\int \frac{9x^3 - 3x + 1}{x^3 - x^2}\,dx$$

Exercise 41: Evaluating Integrals

Directions for Exercises 39–54.
Evaluate the integrals in Exercises 39–54.

Exercise 41

$$\int \frac{\cos y\,dy}{\sin^2 y + \sin y - 6}$$

8.7 Numerical Integration

Assigned: 8, 24, 25, 28, 29, 31

Thomas' Calculus, 14th SI edition • Section 8.7 • textbook pp. 510–512

Exercise 8: Estimating Definite Integrals

Directions for Exercises 1–10.
The instructions for the integrals in Exercises 1–10 have two parts, one for the Trapezoidal Rule and one for Simpson's Rule.

I. Using the Trapezoidal Rule

a. Estimate the integral with $n = 4$ steps and find an upper bound for $|E_T|$.

b. Evaluate the integral directly and find $|E_T|$.

c. Use the formula $(|E_T|/(\text{true value})) \times 100$ to express $|E_T|$ as a percentage of the integral's true value.

II. Using Simpson's Rule

a. Estimate the integral with $n = 4$ steps and find an upper bound for $|E_S|$.

b. Evaluate the integral directly and find $|E_S|$.

c. Use the formula $(|E_S|/(\text{true value})) \times 100$ to express $|E_S|$ as a percentage of the integral's true value.

Exercise 8

$$\int_2^4 \frac{1}{(s-1)^2}\,ds$$

Exercises 24, 25: Estimates with Numerical Data

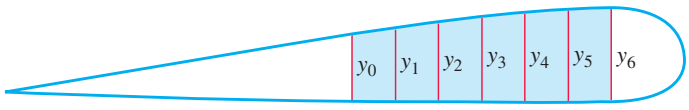
Exercise 24

Distance traveled The accompanying table shows time-to-speed data for a car accelerating from rest to 130 km/h. How far had the car traveled by the time it reached this speed? (Use trapezoids to estimate the area under the velocity curve, but be careful: The time intervals vary in length.)

Speed change	Time (s)
Zero to 30 km/h	2.2
40 km/h	3.2
50 km/h	4.5
60 km/h	5.9
70 km/h	7.8
80 km/h	10.2
90 km/h	12.7
100 km/h	16.0
110 km/h	20.6
120 km/h	26.2
130 km/h	37.1

Exercise 25

Wing design The design of a new airplane requires a gasoline tank of constant cross-sectional area in each wing. A scale drawing of a cross-section is shown here. The tank must hold 2000 kg of gasoline, which has a density of 673 kg/m^3. Estimate the length of the tank by Simpson's Rule.



$y_0 = 0.5$ m, $y_1 = 0.55$ m, $y_2 = 0.6$ m, $y_3 = 0.65$ m,
$y_4 = 0.7$ m, $y_5 = y_6 = 0.75$ m Horizontal spacing $= 0.3$ m

Exercises 28, 29, 31: Theory and Examples

Exercise 28

The error function *The error function*,

$$\text{erf}\,(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt,$$

important in probability and in the theories of heat flow and signal transmission, must be evaluated numerically because there is no elementary expression for the antiderivative of e^{-t^2}.

a. Use Simpson's Rule with $n = 10$ to estimate erf (1).

b. In $[\,0, 1\,]$,

$$\left|\frac{d^4}{dt^4}\left(e^{-t^2}\right)\right| \le 12.$$

Give an upper bound for the magnitude of the error of the estimate in part (a).

Exercise 29

Prove that the sum T in the Trapezoidal Rule for $\int_a^b \boldsymbol{f}(x)\,dx$ is a Riemann sum for $\boldsymbol{f}$ continuous on $[\,a, b\,]$. (*Hint:* Use the Intermediate Value Theorem to show the existence of c_k in the subinterval $[x_{k-1}, x_k]$ satisfying $\boldsymbol{f}(c_k) = (\boldsymbol{f}(x_{k-1}) + \boldsymbol{f}(x_k))/2$.)

Exercise 31

Elliptic integrals The length of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

turns out to be

$$\text{Length} = 4a\int_0^{\pi/2} \sqrt{1 - e^2\cos^2 t}\,dt,$$

where $e = \sqrt{a^2 - b^2}/a$ is the ellipse's eccentricity. The integral in this formula, called an *elliptic integral*, is nonelementary except when $e = 0$ or 1.

a. Use the Trapezoidal Rule with $n = 10$ to estimate the length of the ellipse when $a = 1$ and $e = 1/2$.

b. Use the fact that the absolute value of the second derivative of $\boldsymbol{f}(t) = \sqrt{1 - e^2\cos^2 t}$ is less than 1 to find an upper bound for the error in the estimate you obtained in part (a).

8.8 Improper Integrals

Assigned: 4, 10, 14, 20, 23, 27, 32, 39, 43, 45, 56, 57, 60, 65, 68, 69, 71, 76, 80

Thomas' Calculus, 14th SI edition • Section 8.8 • textbook pp. 521–523

Exercises 4, 10, 14, 20, 23, 27, 32: Evaluating Improper Integrals

Directions for Exercises 1–34.
The integrals in Exercises 1–34 converge. Evaluate the integrals without using tables.

Exercise 4

$$\int_0^4 \frac{dx}{\sqrt{4-x}}$$

Exercise 10

$$\int_{-\infty}^2 \frac{2\,dx}{x^2+4}$$

Exercise 14

$$\int_{-\infty}^{\infty} \frac{x\,dx}{(x^2+4)^{3/2}}$$

Exercise 20

$$\int_0^{\infty} \frac{16\tan^{-1}x}{1+x^2}\,dx$$

Exercise 23

$$\int_{-\infty}^0 e^{-|x|}\,dx$$

Exercise 27

$$\int_0^2 \frac{ds}{\sqrt{4-s^2}}$$

Exercise 32

$$\int_0^2 \frac{dx}{\sqrt{|x-1|}}$$

Exercises 39, 43, 45, 56, 57, 60, 65, 68: Testing for Convergence

Directions for Exercises 35–68.
In Exercises 35–68, use integration, the Direct Comparison Test, or the Limit Comparison Test to test the integrals for convergence. If more than one method applies, use whatever method you prefer.

Exercise 39

$$\int_0^{\pi/2} \tan\theta\,d\theta$$

Exercise 43

$$\int_0^{\ln 2} x^{-2}e^{-1/x}\,dx$$

Exercise 45

$$\int_0^{\pi} \frac{dt}{\sqrt{t}+\sin t}$$

Exercise 56

$$\int_2^{\infty} \frac{dx}{\sqrt{x^2-1}}$$

Exercise 57

$$\int_1^{\infty} \frac{\sqrt{x+1}}{x^2}\,dx$$

Exercise 60

$$\int_{\pi}^{\infty} \frac{1+\sin x}{x^2}\,dx$$

Exercise 65

$$\int_1^{\infty} \frac{1}{\sqrt{e^x-x}}\,dx$$

Exercise 68

$$\int_{-\infty}^{\infty} \frac{dx}{e^x + e^{-x}}$$

Exercises 69, 71, 76, 80: Theory and Examples

Exercise 69

Find the values of p for which each integral converges.

a. $\displaystyle\int_1^2 \frac{dx}{x(\ln x)^p}$ **b.** $\displaystyle\int_2^{\infty} \frac{dx}{x(\ln x)^p}$

Directions for Exercises 71–74.
Exercises 71–74 are about the infinite region in the first quadrant between the curve $y = e^{-x}$ and the x-axis.

Exercise 71

Find the area of the region.

Reference context: Exercise 75.
Find the area of the region that lies between the curves $y = \sec x$ and $y = \tan x$ from $x = 0$ to $x = \pi/2$.

Exercise 76

The region in Exercise 75 is revolved about the x-axis to generate a solid.

a. Find the volume of the solid.

b. Show that the inner and outer surfaces of the solid have infinite area.

Exercise 80

Evaluate $\displaystyle\int_3^{\infty} \frac{dx}{x\sqrt{x^2 - 9}}$.

16.1 Solutions, Slope Fields, and Euler's Method

Assigned: 1–6, 18

Thomas' Calculus, 14th SI edition • Section 16.1 • textbook pp. 1022–1024

Course note: Do Exercise 18 after learning Sections 7.4 and 16.2.

Exercises 1–6: Slope Fields

Directions for Exercises 1–4.
In Exercises 1–4, match the differential equations with their slope fields, graphed here.

Exercise 1

$y' = x + y$

Exercise 2

$y' = y + 1$

Exercise 3

$y' = -\frac{x}{y}$

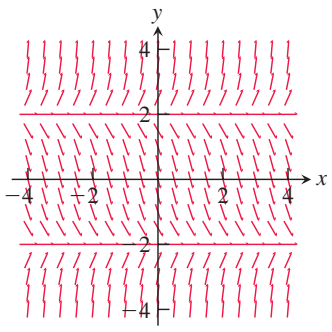
Exercise 4

$y' = y^2 - x^2$

Directions for Exercises 5–6.
In Exercises 5 and 6, copy the slope fields and sketch in some of the solution curves.

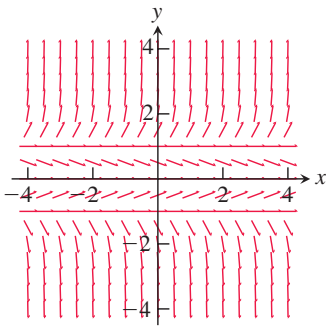
Exercise 5

$y' = (y + 2)(y - 2)$



Exercise 6

$y' = y(y + 1)(y - 1)$



Exercise 18: Using Euler's Method

Directions for Exercises 15–20.
In Exercises 15–20, use Euler's method to calculate the first three approximations to the given initial value problem for the specified increment size. Calculate the exact solution and investigate the accuracy of your approximations. Round your results to four decimal places.

Exercise 18

$y' = y^2(1 + 2x), \quad y(-1) = 1, \quad dx = 0.5$

7.4 Exponential Change and Separable Differential Equations

Assigned: 11–13, 17, 30, 34, 42

Thomas' Calculus, 14th SI edition • Section 7.4 • textbook pp. 423–425

Exercises 11–13, 17: Separable Differential Equations

Directions for Exercises 9–22.
Solve the differential equations in Exercises 9–22.

Exercise 11

$$\frac{dy}{dx} = e^{x-y}$$

Exercise 12

$$\frac{dy}{dx} = 3x^2\, e^{-y}$$

Exercise 13

$$\frac{dy}{dx} = \sqrt{y}\cos^2\sqrt{y}$$

Exercise 17

$$\frac{dy}{dx} = 2x\sqrt{1-y^2}, \quad -1 < y < 1$$

Exercises 30, 34, 42: Applications and Examples

Exercise 30

Growth of bacteria A colony of bacteria is grown under ideal conditions in a laboratory so that the population increases exponentially with time. At the end of 3 hours there are 10,000 bacteria. At the end of 5 hours there are 40,000. How many bacteria were present initially?

Exercise 34

The U.S. population The U.S. Census Bureau keeps a running clock totaling the U.S. population. On September 20, 2012, the total was increasing at the rate of 1 person every 12 s. The population figure for 8:11 p.m. EST on that day was 314,419,198.

a. Assuming exponential growth at a constant rate, find the rate constant for the population's growth (people per 365-day year).

b. At this rate, what will the U.S. population be at 8:11 p.m. EST on September 20, 2019?

Exercise 42

A beam of unknown temperature An aluminum beam was brought from the outside cold into a machine shop where the temperature was held at 18°C. After 10 min, the beam warmed to 2°C and after another 10 min it was 10°C. Use Newton's Law of Cooling to estimate the beam's initial temperature.

16.2 First-Order Linear Equations

Assigned: 1, 4, 16, 17

Thomas' Calculus, 14th SI edition • Section 16.2 • textbook pp. 1028–1030

Exercises 1, 4: First-Order Linear Equations

Directions for Exercises 1–14.
Solve the differential equations in Exercises 1–14.

Exercise 1

$$x\frac{dy}{dx} + y = e^x, \quad x > 0$$

Exercise 4

$$y' + (\tan x)\, y = \cos^2 x, \quad -\pi/2 < x < \pi/2$$

Exercises 16, 17: Solving Initial Value Problems

Directions for Exercises 15–20.
Solve the initial value problems in Exercises 15–20.

Exercise 16

$$t\frac{dy}{dt} + 2y = t^3, \quad t > 0, \quad y(2) = 1$$

Exercise 17

$$\theta\frac{dy}{d\theta} + y = \sin\theta, \quad \theta > 0, \quad y(\pi/2) = 1$$

16.3 Applications

Assigned: 13, 15

Thomas' Calculus, 14th SI edition • Section 16.3 • textbook pp. 1035–1036

Exercises 13, 15: Mixture Problems

Exercise 13

Salt mixture A tank initially contains 400 L of brine in which 20 kg/L of salt are dissolved. A brine containing 0.2 kg/L of salt runs into the tank at the rate of 20 L/min. The mixture is kept uniform by stirring and flows out of the tank at the rate of 16 L/min.

a. At what rate (kilograms per minute) does salt enter the tank at time t?

b. What is the volume of brine in the tank at time t?

c. At what rate (kilograms per minute) does salt leave the tank at time t?

d. Write down and solve the initial value problem describing the mixing process.

e. Find the concentration of salt in the tank 25 min after the process starts.

Exercise 15

Fertilizer mixture A tank contains 400 L of fresh water. A solution containing 0.1 kg/L of soluble lawn fertilizer runs into the tank at the rate of 4 L/min, and the mixture is pumped out of the tank at the rate of 12 L/min. Find the maximum amount of fertilizer in the tank and the time required to reach the maximum.

16.4 Graphical Solutions of Autonomous Equations

Assigned: 1, 7, 10, 12, 14, 18

Thomas' Calculus, 14th SI edition • Section 16.4 • textbook pp. 1042–1043

Course note: No need to study the sign of the second derivative of the solutions.

Exercises 1, 7: Phase Lines and Solution Curves

Directions for Exercises 1–8.
In Exercises 1–8,

- **a.** Identify the equilibrium values. Which are stable and which are unstable?
- **b.** Construct a phase line. Identify the signs of y' and y''.
- **c.** Sketch several solution curves.

Exercise 1

$$\frac{dy}{dx} = (y + 2)(y - 3)$$

Exercise 7

$$y' = (y - 1)(y - 2)(y - 3)$$

Exercises 10, 12, 14: Models of Population Growth

Directions for Exercises 9–12.
The autonomous differential equations in Exercises 9–12 represent models for population growth. For each exercise, use a phase line analysis to sketch solution curves for $P(t)$, selecting different starting values $P(0)$. Which equilibria are stable, and which are unstable?

Exercise 10

$$\frac{dP}{dt} = P(1 - 2P)$$

Exercise 12

$$\frac{dP}{dt} = 3P(1 - P)\left(P - \frac{1}{2}\right)$$

Exercise 14

Controlling a population The fish and game department in a certain state is planning to issue hunting permits to control the deer population (one deer per permit). It is known that if the deer population falls below a certain level m, the deer will become extinct. It is also known that if the deer population rises above the carrying capacity M, the population will decrease back to M through disease and malnutrition.

- **a.** Discuss the reasonableness of the following model for the growth rate of the deer population as a function of time:

$$\frac{dP}{dt} = rP(M - P)(P - m),$$

  where P is the population of the deer and r is a positive constant of proportionality. Include a phase line.
- **b.** Explain how this model differs from the logistic model $dP/dt = rP(M - P)$. Is it better or worse than the logistic model?
- **c.** Show that if $P > M$ for all t, then $\lim_{t\to\infty} P(t) = M$.
- **d.** What happens if $P < m$ for all t?
- **e.** Discuss the solutions to the differential equation. What are the equilibrium points of the model? Explain the dependence of the steady-state value of P on the initial values of P. About how many permits should be issued?

Exercise 18

The spread of information Sociologists recognize a phenomenon called *social diffusion*, which is the spreading of a piece of information, technological innovation, or cultural fad among a population. The members of the population can be divided into two classes: those who have the information and those who do not. In a fixed population whose size is known, it is reasonable to assume that the rate of diffusion is proportional to the number who have the information times the number yet to receive it. If X denotes the number of individuals who have the information in a population of N people, then a mathematical model for social diffusion is given by

$$\frac{dX}{dt} = kX(N - X),$$

where t represents time in days and k is a positive constant.

a. Discuss the reasonableness of the model.

b. Construct a phase line identifying the signs of X' and X''.

c. Sketch representative solution curves.

d. Predict the value of X for which the information is spreading most rapidly. How many people eventually receive the information?